









# 1 Lecture 01: Vector Calculus - Concept of Scalar and Vector Point Function

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 01 of 14 **Video:** <https://www.youtube.com/watch?v=Y4ZFbx4OuAE>

## 1.1 Prerequisites

Before we dive into vector calculus, we must review basic vectors from high school. A scalar is a quantity with only magnitude, like speed or weight. A vector has both magnitude and direction, like velocity.

We represent a point in 3D space with a position vector from the origin  $O$  to a point  $A$ . We call this  $\vec{OA}$ . We define vectors using the unit vectors  $\hat{i}$ ,  $\hat{j}$ , and  $\hat{k}$  along the  $X$ ,  $Y$ , and  $Z$  axes.

We also have two ways to multiply vectors: 1. **Scalar Product (Dot Product):** The result is a scalar number.

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

2. **Vector Product (Cross Product):** The result is a new vector that is perpendicular to both original vectors.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

## 1.2 Scalar Point Functions

Imagine a region  $R$  in 3D space. This is our domain. We can pick any point  $(x, y, z)$  from this space.

A **scalar point function** acts like a magic box. You put a 3D point into the box, and it outputs a single real number (a scalar). It maps a point from 3D space to 1D space.

Here is an example of a scalar point function:

$$f(x, y, z) = xy + x^2z$$

Notice there are no  $\hat{i}$ ,  $\hat{j}$ , or  $\hat{k}$  unit vectors here. Let us evaluate this function at the point  $(1, 2, -1)$ :

$$f(1, 2, -1) = (1)(2) + (1)^2(-1) = 2 - 1 = 1$$

The output is 1, which is a pure scalar number.

## 1.3 Other examples of scalar point functions include:

$$f(x, y, z) = xyz^2$$

•

$$f(x, y, z) = xyz + x^2y^2$$

Physical examples include temperature fields or density fields. At every point in a room, there is a specific temperature number, but no direction.

## 1.4 Vector Point Functions

A **vector point function** is a different kind of magic box. You input a 3D point, and it outputs a vector. It maps a point from 3D space to a new vector in 3D space.

Because the output is a vector, the function rule will always contain the unit vectors  $\hat{i}$ ,  $\hat{j}$ , and  $\hat{k}$ .

Here is an example of a vector point function:

$$\vec{F}(x, y, z) = xy\hat{i} + x^2z\hat{j} + y\hat{k}$$

Let us evaluate this function at the same point  $(1, 2, -1)$ :

$$\vec{F}(1, 2, -1) = (1)(2)\hat{i} + (1)^2(-1)\hat{j} + (2)\hat{k}$$

Simplify the math:

$$\vec{F}(1, 2, -1) = 2\hat{i} - \hat{j} + 2\hat{k}$$

The output is a vector. This vector has both a magnitude and a direction.

### 1.5 Other examples of vector point functions include:

$$\vec{F}(x, y, z) = x^2\hat{i} + yz\hat{j} + z^2\hat{k}$$

•

$$\vec{F}(x, y, z) = x\hat{i} + y^2\hat{j} + xyz\hat{k}$$

Physical examples include velocity fields of fluid flow or gravitational force fields. At every point in space, gravity pulls with a specific strength (magnitude) in a specific direction.

### 1.6 Key Takeaways

- **Scalar Point Function:** Input is a 3D point. Output is a scalar number. The function formula has no unit vectors.
- **Vector Point Function:** Input is a 3D point. Output is a vector. The function formula uses  $\hat{i}$ ,  $\hat{j}$ , and  $\hat{k}$ .
- Vector calculus lets us study these 3D functions just like we study standard 1D functions in normal calculus.

### 1.7 What Comes Next

## 2 Lecture 02: Vector Calculus - Vector Differentiation

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 02 of 14 **Video:** <https://www.youtube.com/watch?v=t-xP272h8Xc>

### 2.1 Prerequisites

### 2.2 Vector Differential Coefficients

When a vector point function  $\vec{F}$  depends on multiple variables like  $x, y$ , and  $z$ , we use partial differentiation. The derivatives we find are called **Vector Differential Coefficients**.

Let us take an example:

$$\vec{F}(x, y, z) = x^2\hat{i} + y^2z\hat{j} + z\hat{k}$$

We can differentiate this partially with respect to  $x$ :

$$\frac{\partial \vec{F}}{\partial x} = 2x\hat{i} + 0\hat{j} + 0\hat{k} = 2x\hat{i}$$

Similarly, with respect to  $y$  and  $z$ :

$$\begin{aligned}\frac{\partial \vec{F}}{\partial y} &= 2yz\hat{j} \\ \frac{\partial \vec{F}}{\partial z} &= y^2\hat{j} + \hat{k}\end{aligned}$$

Notice that the result of differentiating a vector point function is always another vector point function.

### 2.3 Ordinary Vector Differentiation

Often, a vector function depends on a single scalar parameter, like time  $t$ . For example, the position vector  $\vec{r}$  of a moving particle. In this case, we use ordinary differentiation notation:  $\frac{d\vec{r}}{dt}$ .

- **Velocity:** The first derivative of a position vector with respect to time gives the velocity vector  $\vec{v}$ .

$$\vec{v} = \frac{d\vec{r}}{dt}$$

- **Acceleration:** The second derivative of the position vector gives the acceleration vector  $\vec{a}$ .

$$\vec{a} = \frac{d^2\vec{r}}{dt^2}$$

### 2.4 Solved Problems

#### 2.4.1 Problem 1: Velocity and Acceleration

**Question:** If a position vector is given by

$$\vec{r} = a \cos t\hat{i} + a \sin t\hat{j} + t\hat{k},$$

Find the velocity, the acceleration, and the magnitude of the acceleration.

**Solution:** First, find velocity by differentiating once:

$$\vec{v} = \frac{d\vec{r}}{dt} = -a \sin t\hat{i} + a \cos t\hat{j} + \hat{k}$$

Next, find acceleration by differentiating again:

$$\vec{a} = \frac{d^2\vec{r}}{dt^2} = -a \cos t\hat{i} - a \sin t\hat{j}$$

Finally, find the magnitude of the acceleration:

$$\left| \frac{d^2\vec{r}}{dt^2} \right| = \sqrt{(-a \cos t)^2 + (-a \sin t)^2 + 0^2} = \sqrt{a^2(\cos^2 t + \sin^2 t)} = a$$

### 2.4.2 Problem 2: Dot Product Differentiation

**Question:** If

$$\vec{u} = t^2\hat{i} - t\hat{j} + (2t+1)\hat{k}$$

and

$$\vec{v} = (2t-3)\hat{i} + \hat{j} - t\hat{k},$$

Find

$$\frac{d}{dt}(\vec{u} \cdot \vec{v})$$

at  $t = 1$ .

**Solution:** First, compute the dot product algebraically:

$$\vec{u} \cdot \vec{v} = (t^2)(2t-3) + (-t)(1) + (2t+1)(-t)$$

$$\vec{u} \cdot \vec{v} = 2t^3 - 3t^2 - t - 2t^2 - t = 2t^3 - 5t^2 - 2t$$

Now, differentiate the result with respect to  $t$ :

$$\frac{d}{dt}(\vec{u} \cdot \vec{v}) = \frac{d}{dt}(2t^3 - 5t^2 - 2t) = 6t^2 - 10t - 2$$

Finally, evaluate at  $t = 1$ :

$$\left. \frac{d}{dt}(\vec{u} \cdot \vec{v}) \right|_{t=1} = 6(1)^2 - 10(1) - 2 = 6 - 10 - 2 = -6$$

### 2.4.3 Problem 3: Exponential Derivative Identity

**Question:** If

$$\vec{r} = \vec{a}e^{nt} + \vec{b}e^{-nt}$$

, where  $\vec{a}$  and  $\vec{b}$  are constant vectors, prove that

$$\frac{d^2\vec{r}}{dt^2} - n^2\vec{r} = 0.$$

**Solution:** Compute the first derivative. Since  $\vec{a}$  and  $\vec{b}$  are constant, only the exponential terms change:

$$\frac{d\vec{r}}{dt} = n\vec{a}e^{nt} - n\vec{b}e^{-nt}$$

Compute the second derivative:

$$\frac{d^2\vec{r}}{dt^2} = n^2\vec{a}e^{nt} + (-n)^2\vec{b}e^{-nt}$$

$$\frac{d^2\vec{r}}{dt^2} = n^2(\vec{a}e^{nt} + \vec{b}e^{-nt})$$

Notice that the term inside the brackets is our original vector  $\vec{r}$ :

$$\frac{d^2\vec{r}}{dt^2} = n^2\vec{r}$$

Rearranging gives the final proof:

$$\frac{d^2\vec{r}}{dt^2} - n^2\vec{r} = 0$$

### 2.4.4 Problem 4: Position Vector and Kinematics

**Question:** A particle moves so that its position vector is given by

$$\vec{r} = \cos \omega t \hat{i} + \sin \omega t \hat{j},$$

Where  $\omega$  is a constant. Show that the velocity  $\vec{v}$  is perpendicular to  $\vec{r}$ , and that the acceleration  $\vec{a}$  is directed towards the origin.

**Solution:** First, compute velocity:

$$\vec{v} = \frac{d\vec{r}}{dt} = -\omega \sin \omega t \hat{i} + \omega \cos \omega t \hat{j}$$

To check if they are perpendicular, compute the dot product:

$$\vec{v} \cdot \vec{r} = (-\omega \sin \omega t)(\cos \omega t) + (\omega \cos \omega t)(\sin \omega t) = 0$$

Since the dot product is 0, the velocity is perpendicular to the position vector.

Next, compute acceleration:

$$\begin{aligned}\vec{a} &= \frac{d^2\vec{r}}{dt^2} = -\omega^2 \cos \omega t \hat{i} - \omega^2 \sin \omega t \hat{j} \\ \vec{a} &= -\omega^2 (\cos \omega t \hat{i} + \sin \omega t \hat{j}) = -\omega^2 \vec{r}\end{aligned}$$

The negative sign indicates that the acceleration vector points exactly opposite to the position vector. Because the position vector points away from the origin, the acceleration is directed straight towards the origin.

## 2.5 Key Takeaways

- Differentiating a vector point function yields a new vector representing the rate of change.
- If a vector function depends on time  $t$ , its first derivative gives velocity, and its second derivative gives acceleration.
- The standard derivative rules (like product rules and chain rules) apply directly to vector components.

## 2.6 What Comes Next

## 3 Lecture 03: Vector Calculus - Concept of Gradient and Geometric Meaning

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 03 of 14 **Video:** <https://www.youtube.com/watch?v=fqWikeRWBfk>

### 3.1 Prerequisites

We know that a scalar point function  $\phi(x, y, z) = \text{Constant}$  defines a 3D surface. To study how these scalar fields change, we need a special operator that can take derivatives in all spatial directions at once.

### 3.2 The Del Operator

In standard calculus, we use  $\frac{d}{dx}$  to find derivatives. In vector calculus, we use the **Del Operator** (or Nabla), denoted by  $\nabla$ . It is a vector differential operator:

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

Because it contains the unit vectors  $\hat{i}$ ,  $\hat{j}$ , and  $\hat{k}$ , it acts like a vector. We apply this operator to functions to find their rates of change in 3D space.

### 3.3 The Gradient

When we apply the del operator to a **scalar point function**  $\phi$ , we get the **gradient** of  $\phi$ . We write this as  $\text{grad } \phi$  or  $\nabla \phi$ .

$$\nabla \phi = \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

**Important:** The gradient of a scalar point function always results in a **vector quantity**. You input a scalar field, and you output a vector field.

#### 3.3.1 Geometric Meaning

What does this vector physically mean? The gradient  $\nabla \phi$  is the **normal (perpendicular) vector** to the surface defined by  $\phi(x, y, z) = c$  at any given point. If you draw a tangent plane to the surface at a point, the gradient vector points straight out of that plane at a exactly 90 degrees.

### 3.4 Properties of Gradient

The gradient acts very much like a standard derivative. Let  $f$  and  $g$  be two scalar point functions, and  $\vec{a}$  be a constant vector.

1. **Sum/Difference Rule:**

$$\nabla(f \pm g) = \nabla f \pm \nabla g$$

2. **Product Rule:**

$$\nabla(fg) = f\nabla g + g\nabla f$$

3. **Quotient Rule:**

$$\nabla \left( \frac{f}{g} \right) = \frac{g\nabla f - f\nabla g}{g^2}$$

4. **Operator on Position Vector:** If

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k},$$

Then:

$$(\vec{a} \cdot \nabla) \vec{r} = \vec{a}$$

### 3.5 Solved Problems

#### 3.5.1 Problem 1: Basic Gradient Calculation

**Question:** If

$$f = x^3 + y^3 + 3xy$$

, find  $\text{grad } f$ .

**Solution:** Apply the del operator to the function:

$$\text{grad } f = \hat{i} \frac{\partial}{\partial x}(x^3 + y^3 + 3xy) + \hat{j} \frac{\partial}{\partial y}(x^3 + y^3 + 3xy) + \hat{k} \frac{\partial}{\partial z}(x^3 + y^3 + 3xy)$$

Differentiate each part:

$$\text{grad } f = (3x^2 + 3y)\hat{i} + (3y^2 + 3x)\hat{j} + 0\hat{k}$$

Simplify:

$$\text{grad } f = 3(x^2 + y)\hat{i} + 3(y^2 + x)\hat{j}$$

#### 3.5.2 Problem 2: Gradient Evaluation at a Point

**Question:** If

$$\phi(x, y, z) = 2x^2y^3 - 3y^2z^3$$

, find  $\nabla\phi$  at the point  $(1, -1, 1)$ .

**Solution:** First, calculate the partial derivatives:

$$\frac{\partial\phi}{\partial x} = 4xy^3$$

$$\frac{\partial\phi}{\partial y} = 6x^2y^2 - 6yz^3$$

$$\frac{\partial\phi}{\partial z} = -9y^2z^2$$

Combine to form the gradient vector:

$$\nabla\phi = 4xy^3\hat{i} + (6x^2y^2 - 6yz^3)\hat{j} - 9y^2z^2\hat{k}$$

Now, evaluate at  $x = 1$ ,  $y = -1$ ,  $z = 1$ :

$$\left. \frac{\partial\phi}{\partial x} \right|_{(1, -1, 1)} = 4(1)(-1)^3 = -4$$

$$\left. \frac{\partial\phi}{\partial y} \right|_{(1, -1, 1)} = 6(1)^2(-1)^2 - 6(-1)(1)^3 = 6 + 6 = 12$$

$$\left. \frac{\partial\phi}{\partial z} \right|_{(1, -1, 1)} = -9(-1)^2(1)^2 = -9$$

The final vector is:

$$\nabla\phi = -4\hat{i} + 12\hat{j} - 9\hat{k}$$

### 3.5.3 Problem 3: Gradient of a Scalar Triple Product

**Question:** If  $\vec{a}$  and  $\vec{b}$  are constant vectors, prove that

$$\nabla[\vec{r} \cdot \vec{a} \times \vec{b}] = \vec{a} \times \vec{b}.$$

**Solution:** Let the cross product

$$\vec{A} = \vec{a} \times \vec{b}.$$

Since  $\vec{a}$  and  $\vec{b}$  are constant,  $\vec{A}$  is also a constant vector. The scalar triple product becomes a simple dot product:

$$[\vec{r} \cdot \vec{a} \times \vec{b}] = \vec{r} \cdot (\vec{a} \times \vec{b}) = \vec{r} \cdot \vec{A}$$

Let

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

and

$$\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}.$$

Expand the dot product:

$$\vec{r} \cdot \vec{A} = A_1x + A_2y + A_3z$$

Now find the gradient of this scalar:

$$\nabla(\vec{r} \cdot \vec{A}) = \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (A_1x + A_2y + A_3z)$$

$$\nabla(\vec{r} \cdot \vec{A}) = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$$

This perfectly recreates vector  $\vec{A}$ . Therefore:

$$\nabla(\vec{r} \cdot \vec{A}) = \vec{a} \times \vec{b}$$

### 3.5.4 Problem 4: Gradient of

**Question:** If

$$r^2 = x^2 + y^2 + z^2$$

, find  $\text{grad } r^n$ .

**Solution:** First, note that differentiating

$$r^2 = x^2 + y^2 + z^2$$

with respect to  $x$  gives:

$$2r \frac{\partial r}{\partial x} = 2x \implies \frac{\partial r}{\partial x} = \frac{x}{r}$$

Now apply the chain rule to find the gradient of  $r^n$ :

$$\nabla r^n = \hat{i} \frac{\partial(r^n)}{\partial x} + \hat{j} \frac{\partial(r^n)}{\partial y} + \hat{k} \frac{\partial(r^n)}{\partial z}$$

Look at just the  $x$  component:

$$\hat{i} \left( nr^{n-1} \frac{\partial r}{\partial x} \right) = \hat{i} \left( nr^{n-1} \frac{x}{r} \right) = \hat{i} (nxr^{n-2})$$

By symmetry, the  $y$  and  $z$  components will be similar. Adding them all up gives:

$$\nabla r^n = nr^{n-2}(x\hat{i} + y\hat{j} + z\hat{k}) = nr^{n-2}\vec{r}$$

## 3.6 Key Takeaways

- The del operator ( $\nabla$ ) is a vector differential operator.
- The gradient takes a scalar point function and outputs a vector field.
- The gradient vector is geometrically normal (perpendicular) to the original scalar surface.

## 3.7 What Comes Next

## 4 Lecture 04: Vector Calculus - Concept of Directional Derivative

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 04 of 14 **Video:** [https://www.youtube.com/watch?v=UcEozy\\_ig\\_U](https://www.youtube.com/watch?v=UcEozy_ig_U)

### 4.1 Prerequisites

We previously learned that the gradient of a scalar point function  $\phi$ , denoted by  $\nabla\phi$ , gives a vector that is perfectly normal (perpendicular) to the surface. But what if we want to know how the surface changes in an arbitrary direction, not just the normal direction?

### 4.2 Directional Derivative

The **Directional Derivative (D.D.)** is the rate of change of a scalar surface  $\phi(x, y, z)$  at a specific point  $P(x_1, y_1, z_1)$  along the direction of a given vector  $\vec{a}$ .

To calculate it, we evaluate the gradient of  $\phi$  at the given point, and then project it onto the desired direction by taking the dot product with the unit vector  $\hat{a}$ .

$$\text{D.D.} = \frac{d\phi}{ds} = \hat{a} \cdot \nabla\phi$$

Where: -  $\nabla\phi$  is the gradient vector evaluated at point  $P$ . -

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

is the unit vector in the direction of vector  $\vec{a}$ .

Because the dot product yields a scalar, the directional derivative is a scalar magnitude.

#### 4.2.1 Maximum Directional Derivative

The dot product  $\hat{a} \cdot \nabla\phi$  equals

$$|\nabla\phi| \cos \theta$$

. This value is maximized when  $\cos \theta = 1$ , which occurs when  $\theta = 0$ .

This means the directional derivative is **maximum in the exact direction of the normal vector** ( $\nabla\phi$ ). The maximum possible magnitude of the directional derivative is simply the magnitude of the gradient:

$$\text{Max D.D.} = |\nabla\phi|$$

#### 4.2.2 Unit Normal Vector

Since the gradient  $\nabla\phi$  provides a vector normal to the surface, we can easily find the **unit normal vector**  $\hat{n}$  by dividing the gradient by its own magnitude:

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$$

### 4.3 Solved Problems

#### 4.3.1 Problem 1: Basic Directional Derivative

**Question:** Find the directional derivative of

$$\phi = (x^2 + y^2 + z^2)^{-1/2}$$

at the point  $P(3, 1, 2)$  in the direction of the vector

$$\vec{a} = yz\hat{i} + zx\hat{j} + xy\hat{k}.$$

**Solution:** First, compute the gradient  $\nabla\phi$ :

$$\frac{\partial\phi}{\partial x} = -\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2}(2x) = -x(x^2 + y^2 + z^2)^{-3/2}$$

By symmetry, we get similar terms for  $y$  and  $z$ .

$$\nabla\phi = -(x^2 + y^2 + z^2)^{-3/2}(x\hat{i} + y\hat{j} + z\hat{k})$$

Evaluate the gradient at  $P(3, 1, 2)$ :

$$x^2 + y^2 + z^2 = 3^2 + 1^2 + 2^2 = 14$$

$$\nabla\phi|_{(3,1,2)} = \frac{-(3\hat{i} + \hat{j} + 2\hat{k})}{14\sqrt{14}}$$

Next, find the unit vector  $\hat{a}$  along  $\vec{a}$  at the point  $P(3, 1, 2)$ :

$$\vec{a}|_{(3,1,2)} = (1)(2)\hat{i} + (2)(3)\hat{j} + (3)(1)\hat{k} = 2\hat{i} + 6\hat{j} + 3\hat{k}$$

$$|\vec{a}| = \sqrt{2^2 + 6^2 + 3^2} = \sqrt{49} = 7 \implies \hat{a} = \frac{2\hat{i} + 6\hat{j} + 3\hat{k}}{7}$$

Finally, take the dot product  $\hat{a} \cdot \nabla\phi$ :

$$\begin{aligned} \text{D.D.} &= \left( \frac{2\hat{i} + 6\hat{j} + 3\hat{k}}{7} \right) \cdot \left( \frac{-(3\hat{i} + \hat{j} + 2\hat{k})}{14\sqrt{14}} \right) \\ \text{D.D.} &= \frac{-[2(3) + 6(1) + 3(2)]}{98\sqrt{14}} = \frac{-18}{98\sqrt{14}} = -\frac{9}{49\sqrt{14}} \end{aligned}$$

#### 4.3.2 Problem 2: Direction Defined by Points

**Question:** Find the directional derivative of

$$\phi = x^2 - y^2 + 2z^2$$

at the point  $P(1, 2, 3)$  in the direction of the line  $PQ$ , where  $Q$  is the point  $(5, 0, 4)$ .

**Solution:** First, compute the gradient and evaluate it at  $P(1, 2, 3)$ :

$$\nabla\phi = 2x\hat{i} - 2y\hat{j} + 4z\hat{k}$$

$$\nabla\phi|_{(1,2,3)} = 2(1)\hat{i} - 2(2)\hat{j} + 4(3)\hat{k} = 2\hat{i} - 4\hat{j} + 12\hat{k}$$

Next, construct the direction vector  $\vec{PQ}$ :

$$\vec{PQ} = \vec{OQ} - \vec{OP} = (5\hat{i} + 0\hat{j} + 4\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) = 4\hat{i} - 2\hat{j} + \hat{k}$$

Find the unit vector  $\hat{a}$ :

$$|\vec{PQ}| = \sqrt{16 + 4 + 1} = \sqrt{21} \implies \hat{a} = \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{21}}$$

Compute the directional derivative  $\hat{a} \cdot \nabla\phi$ :

$$\text{D.D.} = \left( \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{21}} \right) \cdot (2\hat{i} - 4\hat{j} + 12\hat{k}) = \frac{8 + 8 + 12}{\sqrt{21}} = \frac{28}{\sqrt{21}}$$

### 4.3.3 Problem 3: Finding Unknown Constants

**Question:** Find the constants  $a, b, c$  so that the directional derivative of

$$\phi = axy^2 + byz + cz^2x^3$$

at the point  $(1, 2, -1)$  has a maximum magnitude of 64 in the direction parallel to the  $Z$ -axis.

**Solution:** First, compute the gradient and evaluate at  $(1, 2, -1)$ :

$$\nabla\phi = (ay^2 + 3cz^2x^2)\hat{i} + (2axy + bz)\hat{j} + (by + 2czx^3)\hat{k}$$

$$\nabla\phi|_{(1,2,-1)} = (4a + 3c)\hat{i} + (4a - b)\hat{j} + (2b - 2c)\hat{k}$$

The directional derivative is maximum in the direction of the gradient. For this maximum to occur parallel to the  $Z$ -axis (the  $\hat{k}$  direction), the  $\hat{i}$  and  $\hat{j}$  components must be zero: 1.

$$4a + 3c = 0 \implies c = -\frac{4}{3}a$$

$$2. \quad 4a - b = 0 \implies b = 4a$$

Also, the magnitude of the gradient must equal the maximum magnitude of 64. Since only the  $\hat{k}$  component survives, it must be positive and equal to 64:

$$2b - 2c = 64 \implies b - c = 32$$

Substitute  $b$  and  $c$  in terms of  $a$ :

$$4a - \left(-\frac{4}{3}a\right) = 32 \implies \frac{16}{3}a = 32 \implies a = 6$$

Now, substitute  $a = 6$  back to find  $b$  and  $c$ :

$$b = 4(6) = 24$$

$$c = -\frac{4}{3}(6) = -8$$

Final Answer:  $a = 6, b = 24, c = -8$ .

### 4.3.4 Problem 4: Angle Between Surfaces

**Question:** Find the angle between the surfaces

$$x^2 + y^2 + z^2 = 9$$

and

$$z = x^2 + y^2 - 3$$

at the point  $(2, -1, 2)$ .

**Solution:** The angle between two surfaces is the angle between their normal vectors. Let's find the gradient of both surfaces.

$$\phi_1 = x^2 + y^2 + z^2 - 9 \implies \nabla\phi_1 = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$\phi_2 = x^2 + y^2 - z - 3 \implies \nabla\phi_2 = 2x\hat{i} + 2y\hat{j} - \hat{k}$$

Evaluate both gradients at  $(2, -1, 2)$  to find the normal vectors  $\vec{n}_1$  and  $\vec{n}_2$ :

$$\vec{n}_1 = 4\hat{i} - 2\hat{j} + 4\hat{k}$$

$$\vec{n}_2 = 4\hat{i} - 2\hat{j} - \hat{k}$$

Use the dot product formula

$$\cos \theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$

to find the angle:

$$\begin{aligned} |\vec{n}_1| &= \sqrt{16 + 4 + 16} = 6 \\ |\vec{n}_2| &= \sqrt{16 + 4 + 1} = \sqrt{21} \\ \vec{n}_1 \cdot \vec{n}_2 &= (4)(4) + (-2)(-2) + (4)(-1) = 16 + 4 - 4 = 16 \\ \theta &= \cos^{-1} \left( \frac{16}{6\sqrt{21}} \right) = \cos^{-1} \left( \frac{8}{3\sqrt{21}} \right) \end{aligned}$$

### 4.3.5 Problem 5: Combined Concepts

**Question:** For the surface

$$\phi = x^2yz + 4xz^2$$

at the point  $(1, -2, -1)$ : 1. Find the directional derivative in the direction

$$2\hat{i} - \hat{j} - 2\hat{k}.$$

2. In what direction is the directional derivative maximum, and what is its magnitude?
3. Find the unit normal vector to the surface.

**Solution:** First, evaluate the gradient at  $(1, -2, -1)$ :

$$\nabla \phi = (2xyz + 4z^2)\hat{i} + (x^2z)\hat{j} + (x^2y + 8xz)\hat{k}$$

$$\nabla \phi|_{(1, -2, -1)} = 8\hat{i} - \hat{j} - 10\hat{k}$$

**Part 1: Directional Derivative** Find unit vector  $\hat{a}$  for

$$2\hat{i} - \hat{j} - 2\hat{k} ::$$

$$\hat{a} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}$$

$$\text{D.D.} = \hat{a} \cdot \nabla \phi = \left( \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3} \right) \cdot (8\hat{i} - \hat{j} - 10\hat{k}) = \frac{16 + 1 + 20}{3} = \frac{37}{3}$$

**Part 2: Maximum Directional Derivative** The maximum directional derivative is in the direction of the gradient itself:

$$8\hat{i} - \hat{j} - 10\hat{k}.$$

Its magnitude is  $|\nabla \phi|$ :

$$\text{Max D.D.} = \sqrt{8^2 + (-1)^2 + (-10)^2} = \sqrt{64 + 1 + 100} = \sqrt{165}$$

**Part 3: Unit Normal Vector** Divide the gradient by its magnitude to find the unit normal vector:

$$\hat{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{8\hat{i} - \hat{j} - 10\hat{k}}{\sqrt{165}}$$

## 4.4 Key Takeaways

- The directional derivative measures the rate of change of a surface along a chosen vector direction.
- It is calculated by taking the dot product of the surface's gradient and the unit vector of the desired direction ( $\hat{a} \cdot \nabla \phi$ ).
- The directional derivative is maximum in the direction of the normal vector (the gradient), and its maximum value is exactly the magnitude of the gradient ( $|\nabla \phi|$ ).

## 4.5 What Comes Next

We have seen how the del operator  $\nabla$  applies to a scalar point function to produce a gradient vector. Next, we will see what happens when we apply the del operator to a vector point function. This introduces the concepts of **Divergence** (dot product) and **Curl** (cross product).

## 5 Lecture 05: Vector Calculus - Concept of Curl and Divergence

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 05 of 14 **Video:** <https://www.youtube.com/watch?v=Zdd8Tf5tkvM>

### 5.1 Prerequisites

We have seen that applying the Del operator ( $\nabla$ ) to a scalar function produces the gradient (a vector). In this lecture, we apply the Del operator to a **vector point function**

$$\vec{F}(x, y, z) = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}.$$

Because it is a vector operator acting on a vector function, we can do this in two ways: via a dot product (Divergence) or a cross product (Curl).

### 5.2 Divergence of a Vector

The **divergence** of a vector field  $\vec{F}$  is defined as the dot product of the Del operator and  $\vec{F}$ . It is written as  $\text{div } \vec{F}$  or  $\nabla \cdot \vec{F}$ .

$$\begin{aligned}\text{div } \vec{F} &= \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (F_1\hat{i} + F_2\hat{j} + F_3\hat{k}) \\ \text{div } \vec{F} &= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}\end{aligned}$$

Because it is a dot product, the divergence is a **scalar quantity**. Physically, it represents the net flux per unit volume emitting from a point (e.g., heat radiating from a source, or fluid expanding outward).

#### 5.2.1 Solenoidal Vectors

If the divergence of a vector field is zero everywhere (

$$\nabla \cdot \vec{F} = 0$$

), the vector is called **solenoidal**. This means there is no net expansion or compression. It behaves like an incompressible fluid with no sources or sinks.

### 5.3 Curl of a Vector

The **curl** of a vector field  $\vec{F}$  is defined as the cross product of the Del operator and  $\vec{F}$ . It is written as “curl  $\vec{F}$ ” or  $\nabla \times \vec{F}$ .

We compute this using the standard cross product determinant:

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

Expanding this gives:

$$\text{curl } \vec{F} = \hat{i} \left( \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) - \hat{j} \left( \frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z} \right) + \hat{k} \left( \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)$$

Because it is a cross product, the curl is a **vector quantity**. Physically, it measures the rotational tendency or “spin” of the vector field at a point (like a vortex or whirlpool).

### 5.3.1 Irrotational Vectors

If the curl of a vector field is the zero vector everywhere, i.e.

$$\nabla \times \vec{F} = \vec{0}$$

the vector is called **irrotational**. This means the field has absolutely no rotational flow or twisting motion.

## 5.4 Important Vector Identities

There are two critical identities involving these operations: 1. **Divergence of Curl is Zero:** The divergence of the curl of any vector field is always exactly zero.

$$\text{div} (\text{curl } \vec{F}) = \nabla \cdot (\nabla \times \vec{F}) = 0$$

2. **Curl of Gradient is Zero:** The curl of the gradient of any scalar field is always the zero vector.

$$\text{curl} (\text{grad } \phi) = \nabla \times (\nabla \phi) = \vec{0}$$

## 5.5 Solved Problems

### 5.5.1 Problem 1: Solenoidal Condition

**Question:** Determine  $a$  such that the vector

$$\vec{F} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x - az)\hat{k}$$

is solenoidal.

**Solution:** Since the vector is solenoidal,

$$\nabla \cdot \vec{F} = 0.$$

$$\frac{\partial}{\partial x}(x + 3y) + \frac{\partial}{\partial y}(y - 2z) + \frac{\partial}{\partial z}(x - az) = 0$$

$$1 + 1 - a = 0 \implies a = 2$$

### 5.5.2 Problem 2: Divergence at a Point

**Question:** If

$$\vec{F} = xy \sin z \hat{i} + y^2 \sin x \hat{j} + z^2 \sin xy \hat{k},$$

Find  $\text{div } \vec{F}$  at the point  $(0, \pi/2, \pi/2)$ .

**Solution:** First, compute the divergence:

$$\text{div } \vec{F} = \frac{\partial}{\partial x}(xy \sin z) + \frac{\partial}{\partial y}(y^2 \sin x) + \frac{\partial}{\partial z}(z^2 \sin xy)$$

$$\text{div } \vec{F} = y \sin z + 2y \sin x + 2z \sin xy$$

Substitute  $x = 0, y = \pi/2, z = \pi/2$ :

$$\text{div } \vec{F} = \left(\frac{\pi}{2}\right) \sin\left(\frac{\pi}{2}\right) + 2\left(\frac{\pi}{2}\right) \sin(0) + 2\left(\frac{\pi}{2}\right) \sin(0)$$

$$\text{div } \vec{F} = \frac{\pi}{2}(1) + 0 + 0 = \frac{\pi}{2}$$

### 5.5.3 Problem 3: Properties of Position Vector

**Question:** If

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k},$$

Prove that

$$\text{div } \vec{r} = 3$$

and

$$\text{curl } \vec{r} = \vec{0}.$$

**Solution:** For divergence:

$$\nabla \cdot \vec{r} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$$

For curl:

$$\nabla \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix}$$

Evaluating the determinant gives

$$\hat{i}(0 - 0) - \hat{j}(0 - 0) + \hat{k}(0 - 0) = \vec{0}.$$

Both properties are proven.

### 5.5.4 Problem 4: Evaluating Irrotational Flow

**Question:** Show that

$$\vec{F} = (\sin y + z)\hat{i} + (x \cos y - z)\hat{j} + (x - y)\hat{k}$$

is irrotational.

**Solution:** Compute the curl:

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin y + z & x \cos y - z & x - y \end{vmatrix}$$

Evaluate the  $\hat{i}$  component:

$$\frac{\partial}{\partial y}(x - y) - \frac{\partial}{\partial z}(x \cos y - z) = -1 - (-1) = 0$$

Evaluate the  $\hat{j}$  component:

$$\frac{\partial}{\partial z}(\sin y + z) - \frac{\partial}{\partial x}(x - y) = 1 - 1 = 0$$

Evaluate the  $\hat{k}$  component:

$$\frac{\partial}{\partial x}(x \cos y - z) - \frac{\partial}{\partial y}(\sin y + z) = \cos y - \cos y = 0$$

Since

$$\nabla \times \vec{F} = 0\hat{i} + 0\hat{j} + 0\hat{k} = \vec{0},$$

The vector is irrotational.

### 5.5.5 Problem 5: Curl of a Cross Product Identity

**Question:** If  $\vec{a}$  is a constant vector and  $\vec{r}$  is the position vector, prove that

$$\text{curl } (\vec{r} \times \vec{a}) = -2\vec{a}.$$

**Solution:** Use the vector identity:

$$\nabla \times (\vec{A} \times \vec{B}) = \vec{A}(\nabla \cdot \vec{B}) - \vec{B}(\nabla \cdot \vec{A}) + (\vec{B} \cdot \nabla)\vec{A} - (\vec{A} \cdot \nabla)\vec{B}$$

Let  $A = r$  and  $B = a$ .

- Since  $a$  is constant:

$$\nabla \cdot a = 0$$

and

$$(r \cdot \nabla)a = 0$$

- From earlier:

$$\nabla \cdot r = 3$$

- 

$$(a \cdot \nabla)r = a$$

Substitute these into the identity:

$$\text{curl } (\vec{r} \times \vec{a}) = \vec{r}(0) - \vec{a}(3) + \vec{a} - \vec{0} = -3\vec{a} + \vec{a} = -2\vec{a}$$

### 5.5.6 Problem 6: Operations on

**Question:** Prove that

$$\text{div}(r^n \vec{r}) = (n+3)r^n.$$

For what value of  $n$  is it solenoidal?

**Solution:** First, note that

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

and similarly for  $y$  and  $z$ . Compute the divergence:

$$\nabla \cdot (r^n x \hat{i} + r^n y \hat{j} + r^n z \hat{k}) = \frac{\partial}{\partial x}(r^n x) + \frac{\partial}{\partial y}(r^n y) + \frac{\partial}{\partial z}(r^n z)$$

Apply the product rule to the  $x$  term:

$$\frac{\partial}{\partial x}(r^n x) = r^n(1) + x \left( n r^{n-1} \frac{\partial r}{\partial x} \right) = r^n + n x r^{n-1} \left( \frac{x}{r} \right) = r^n + n x^2 r^{n-2}$$

Summing the terms for  $x, y, z$ :

$$\text{div}(r^n \vec{r}) = 3r^n + n r^{n-2}(x^2 + y^2 + z^2)$$

Since

$$x^2 + y^2 + z^2 = r^2 :$$

$$\text{div}(r^n \vec{r}) = 3r^n + n r^{n-2}(r^2) = 3r^n + n r^n = (n+3)r^n$$

For the vector to be solenoidal, the divergence must be zero:

$$(n+3)r^n = 0 \implies n = -3$$

## 5.6 Key Takeaways

- **Divergence** ( $\nabla \cdot \vec{F}$ ) is a dot product that measures the outward flux or expansion of a vector field (scalar output).
- **Curl** ( $\nabla \times \vec{F}$ ) is a cross product that measures the rotation or spin of a vector field (vector output).
- A field with zero divergence is **solenoidal**; a field with zero curl is **irrotational**.

## 5.7 What Comes Next

We have completed our study of vector differential operators (Gradient, Divergence, and Curl). The next major topic in vector calculus is **Vector Integration**, starting with Line Integrals and calculating work done by a force.

## 6 Lecture 06: Vector Calculus - Concept of Vector Integration and Line Integration

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 06 of 14 **Video:** <https://www.youtube.com/watch?v=0CRFIOOSA9c>

### 6.1 Prerequisites

We have thoroughly covered vector differentiation (Gradient, Divergence, and Curl). Now we move in the opposite direction: **Vector Integration**. There are three main types of integrals in vector calculus: 1. **Line Integral** 2. **Surface Integral** 3. **Volume Integral**

A solid understanding of these integrals forms the foundation for three major theorems: Green's Theorem, Stokes' Theorem, and Gauss Divergence Theorem. We will start with standard vector integration and Line Integrals.

### 6.2 1. Integration of Vector Functions

Just like scalar functions, a vector point function

$$\vec{F}(t) = F_1(t)\hat{i} + F_2(t)\hat{j} + F_3(t)\hat{k}$$

depending on a single scalar variable  $t$  can be integrated.

#### 6.2.1 Indefinite Integral

To integrate indefinitely, integrate each component separately and add an arbitrary constant vector

$$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}.$$
$$\int \vec{F}(t)dt = \left(\int F_1(t)dt\right)\hat{i} + \left(\int F_2(t)dt\right)\hat{j} + \left(\int F_3(t)dt\right)\hat{k} + \vec{c}$$

#### 6.2.2 Definite Integral

To integrate over an interval  $[a, b]$ , compute the integral and apply the upper and lower limits to each component.

$$\int_a^b \vec{F}(t)dt = \left[\vec{f}(t)\right]_a^b = \vec{f}(b) - \vec{f}(a)$$

### 6.3 2. Line Integration

A line integral computes the integral of a vector field  $\vec{F}$  along a specific curve  $C$  in space. Let

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

be the position vector of any point on curve  $C$ . The differential displacement vector is

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}.$$

The line integral of  $\vec{F}$  along curve  $C$  is defined by the dot product:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (F_1dx + F_2dy + F_3dz)$$

### 6.3.1 How to Evaluate Line Integrals

You cannot integrate directly when  $x, y$ , and  $z$  are all mixed together. **The curve  $C$  provides the relationship between the variables.** The most reliable method is to use parametric equations. If the curve  $C$  is described by  $x = x(t)$ ,  $y = y(t)$ , and  $z = z(t)$  as  $t$  varies from  $t_1$  to  $t_2$ :

$$\int_C \vec{F} \cdot d\vec{r} = \int_{t_1}^{t_2} \left( F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt$$

If  $C$  is a closed curve (like a circle or rectangle), the line integral is called the **circulation** of  $\vec{F}$ , denoted as  $\oint_C \vec{F} \cdot d\vec{r}$ . By standard convention, closed paths are traversed in the anti-clockwise direction.

## 6.4 Solved Problems

### 6.4.1 Problem 1: Basic Definite Integral

**Question:** If

$$\vec{F}(t) = (t - t^2)\hat{i} + 2t^3\hat{j} - 3\hat{k},$$

Find  $\int_1^2 \vec{F}(t) dt$ .

**Solution:** Integrate each component with respect to  $t$ :

$$\int_1^2 \left[ (t - t^2)\hat{i} + 2t^3\hat{j} - 3\hat{k} \right] dt = \left[ \left( \frac{t^2}{2} - \frac{t^3}{3} \right)\hat{i} + \frac{t^4}{2}\hat{j} - 3t\hat{k} \right]_1^2$$

Apply the upper limit ( $t = 2$ ):

$$\left( \frac{4}{2} - \frac{8}{3} \right)\hat{i} + \frac{16}{2}\hat{j} - 6\hat{k} = -\frac{2}{3}\hat{i} + 8\hat{j} - 6\hat{k}$$

Apply the lower limit ( $t = 1$ ):

$$\left( \frac{1}{2} - \frac{1}{3} \right)\hat{i} + \frac{1}{2}\hat{j} - 3\hat{k} = \frac{1}{6}\hat{i} + \frac{1}{2}\hat{j} - 3\hat{k}$$

Subtract the two:

$$\left( -\frac{2}{3} - \frac{1}{6} \right)\hat{i} + \left( 8 - \frac{1}{2} \right)\hat{j} + (-6 - (-3))\hat{k} = -\frac{5}{6}\hat{i} + \frac{15}{2}\hat{j} - 3\hat{k}$$

### 6.4.2 Problem 2: Finding Velocity and Displacement

**Question:** The acceleration of a particle is

$$\vec{a} = 12 \cos 2t\hat{i} - 8 \sin 2t\hat{j} + 16t\hat{k}.$$

Find velocity  $\vec{v}$  and displacement  $\vec{r}$  given that at  $t = 0$ ,

$$\vec{v} = 0$$

and

$$\vec{r} = 0.$$

**Solution:** Integrate acceleration to find velocity  $\vec{v}(t)$ :

$$\vec{v}(t) = \int \vec{a}(t) dt = 6 \sin 2t\hat{i} + 4 \cos 2t\hat{j} + 8t^2\hat{k} + \vec{c}_1$$

Apply boundary condition

$$\vec{v}(0) = 0 :$$

$$0 = 0\hat{i} + 4\hat{j} + 0\hat{k} + \vec{c}_1 \implies \vec{c}_1 = -4\hat{j}$$

$$\vec{v}(t) = 6 \sin 2t \hat{i} + (4 \cos 2t - 4) \hat{j} + 8t^2 \hat{k}$$

Integrate velocity to find displacement  $\vec{r}(t)$ :

$$\vec{r}(t) = \int \vec{v}(t) dt = -3 \cos 2t \hat{i} + (2 \sin 2t - 4t) \hat{j} + \frac{8}{3} t^3 \hat{k} + \vec{c}_2$$

Apply boundary condition

$$\vec{r}(0) = 0 :$$

$$0 = -3\hat{i} + 0\hat{j} + 0\hat{k} + \vec{c}_2 \implies \vec{c}_2 = 3\hat{i}$$

$$\vec{r}(t) = (3 - 3 \cos 2t) \hat{i} + (2 \sin 2t - 4t) \hat{j} + \frac{8}{3} t^3 \hat{k}$$

### 6.4.3 Problem 3: Line Integration on a Parametric Curve

**Question:** Evaluate  $\int_C \vec{F} \cdot d\vec{r}$  where

$$\vec{F} = xy\hat{i} + yz\hat{j} + zx\hat{k}$$

and  $C$  is

$$\vec{r} = t\hat{i} + t^2\hat{j} + t^3\hat{k}$$

from  $t = -1$  to  $1$ .

**Solution:** From the curve equation,

$$x = t, y = t^2, z = t^3.$$

Differentiating gives

$$dx = dt, dy = 2t dt, dz = 3t^2 dt.$$

Substitute  $x, y, z$  into  $\vec{F}$ :

$$\vec{F} = (t)(t^2)\hat{i} + (t^2)(t^3)\hat{j} + (t^3)(t)\hat{k} = t^3\hat{i} + t^5\hat{j} + t^4\hat{k}$$

The displacement vector is

$$d\vec{r} = (\hat{i} + 2t\hat{j} + 3t^2\hat{k})dt.$$

Take the dot product  $\vec{F} \cdot d\vec{r}$ :

$$\vec{F} \cdot d\vec{r} = (t^3)(1) + (t^5)(2t) + (t^4)(3t^2) = (t^3 + 2t^6 + 3t^6)dt = (t^3 + 5t^6)dt$$

Integrate from  $-1$  to  $1$ :

$$\begin{aligned} \int_{-1}^1 (t^3 + 5t^6) dt &= \left[ \frac{t^4}{4} + \frac{5t^7}{7} \right]_{-1}^1 \\ &= \left( \frac{1}{4} + \frac{5}{7} \right) - \left( \frac{1}{4} - \frac{5}{7} \right) = \frac{10}{7} \end{aligned}$$

### 6.4.4 Problem 4: Line Integration on a Parabola

**Question:** Evaluate  $\int_C \vec{F} \cdot d\vec{r}$  where

$$\vec{F} = x^2 y^2 \hat{i} + y \hat{j}$$

and  $C$  is the parabola

$$y^2 = 4x$$

from  $(0, 0)$  to  $(4, 4)$ .

**Solution:** The integrand is

$$\vec{F} \cdot d\vec{r} = x^2 y^2 dx + y dy.$$

Using the curve equation

$$y^2 = 4x$$

, we substitute it into the  $dx$  term. From the coordinates, both  $x$  and  $y$  vary from 0 to 4.

$$\begin{aligned}\int_C \vec{F} \cdot d\vec{r} &= \int_0^4 x^2(4x)dx + \int_0^4 ydy = \int_0^4 4x^3dx + \int_0^4 ydy \\ &= [x^4]_0^4 + \left[\frac{y^2}{2}\right]_0^4 = (256) + \left(\frac{16}{2}\right) = 256 + 8 = 264\end{aligned}$$

#### 6.4.5 Problem 5: Line Integration on an Arbitrary Path

**Question:** Evaluate

$$\int_C [yzdx + (zx + 1)dy + xydz]$$

where  $C$  is any path from  $(1, 0, 0)$  to  $(2, 1, 4)$ .

**Solution:** Since the integral is path-independent (it specifies “any path”), we choose the straight line connecting the two points. The equation of the line is

$$\frac{x-1}{2-1} = \frac{y-0}{1-0} = \frac{z-0}{4-0} = t.$$

This gives parametric equations:  $x = t + 1, y = t, z = 4t$ . Differentiating gives:  $dx = dt, dy = dt, dz = 4dt$ . At  $(1, 0, 0)$ ,  $t = 0$ . At  $(2, 1, 4)$ ,  $t = 1$ . Substitute these into the integrand:

$$(t)(4t)dt + ((4t)(t+1) + 1)dt + (t+1)(t)(4dt) = (4t^2 + 4t^2 + 4t + 1 + 4t^2 + 4t)dt = (12t^2 + 8t + 1)dt$$

Integrate from 0 to 1:

$$\int_0^1 (12t^2 + 8t + 1)dt = [4t^3 + 4t^2 + t]_0^1 = 4(1) + 4(1) + 1 = 9$$

### 6.5 Key Takeaways

- The curve  $C$  provides the critical constraints (relationship between variables) needed to evaluate a line integral.
- By substituting the curve’s parametric equations into the line integral

$$\int_C (F_1dx + F_2dy + F_3dz),$$

You can convert the entire integral into a single variable  $t$ . \* For line integrals over segmented paths (like a rectangle or triangle), split the integral into a sum of integrals for each segment, defining the unique bounds and constant variables for each.

### 6.6 What Comes Next

We have seen how to calculate the integral over a one-dimensional path (Line Integration). Next, we will expand this into two dimensions with **Surface Integration**, where we will integrate over a 2D surface in 3D space.

## 7 Lecture 07: Vector Calculus - Concept of Surface and Volume Integral (Part 1)

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 07 of 14 **Video:** <https://www.youtube.com/watch?v=5BLWsAa3Yr8>

### 7.1 Prerequisites

We have completed our study of Line Integrals, which involve integrating a vector field over a one-dimensional path (a curve). Now, we extend our integration to two dimensions (Surface Integrals) and three dimensions (Volume Integrals).

### 7.2 1. Surface Integration

A surface integral computes the integral of a vector field over a 3D surface  $S$ . It is equivalent to a double integral in standard calculus, as it requires two parameters to define a surface.

The surface integral of a vector point function  $\mathbf{F}$  over a surface  $S$  is given by:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS$$

Where: -  $\mathbf{F}$  is the vector field. -  $\hat{\mathbf{n}}$  is the unit normal vector pointing outward from the surface  $S$ . -  $dS$  is the infinitesimal surface area element of  $S$ .

#### 7.2.1 Finding the Unit Normal Vector

If the surface  $S$  is represented by a scalar equation  $\phi(x, y, z) = c$ , then the unit normal vector  $\hat{\mathbf{n}}$  is simply the normalized gradient:

$$\hat{\mathbf{n}} = \frac{\nabla \phi}{|\nabla \phi|}$$

#### 7.2.2 Projecting the Surface

Because the surface  $S$  exists in 3D space, we cannot integrate  $dS$  directly. We must **project** the 3D surface onto one of the 2D coordinate planes ( $xy$ ,  $yz$ , or  $zx$ ). This allows us to establish the limits of integration for the two projected variables.

The formulas for the projected area element  $dS$  are: 1. **Projection on  $xy$ -plane:**

$$dS = \frac{dx dy}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}|}$$

2. **Projection on  $yz$ -plane:**

$$dS = \frac{dy dz}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{i}}|}$$

3. **Projection on  $zx$ -plane:**

$$dS = \frac{dz dx}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{j}}|}$$

*Rule of thumb: The denominator is the dot product of  $\hat{\mathbf{n}}$  and the unit vector of the axis orthogonal to the projection plane.*

## 7.3 2. Volume Integration

Volume integration extends the concept to a three-dimensional region (or volume)  $V$ . It is represented as a triple integral:

$$\iiint_V \mathbf{F} dV$$

In Cartesian coordinates, the volume element is simply  $dV = dx dy dz$ .

### 7.3.1 Setting Limits

To find the bounds for volume integration: 1. Project the 3D volume onto a 2D coordinate plane (e.g., the  $xy$ -plane). The boundary of this 2D shadow provides the limits for two variables (e.g.,  $x$  and  $y$ ). 2. The limits for the third variable (e.g.,  $z$ ) are defined by the “floor” and “ceiling” surfaces of the volume along that axis.

## 7.4 Solved Problems

### 7.4.1 Problem 1: Surface Integral Over a Plane

**Question:** If

$$\mathbf{F} = 6z\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + y\hat{\mathbf{k}},$$

Evaluate

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS,$$

Where  $S$  is the portion of the plane  $2x + 3y + 6z = 12$  which lies in the first octant.

**Solution: Step 1: Understand the Surface  $S$**  The plane  $2x + 3y + 6z = 12$  in the first octant ( $x, y, z > 0$ ) forms a triangular surface. We find its intercepts by setting two variables to 0: -  $x$ -intercept:  $(6, 0, 0)$  -  $y$ -intercept:  $(0, 4, 0)$  -  $z$ -intercept:  $(0, 0, 2)$

**Step 2: Find the Unit Normal  $\hat{\mathbf{n}}$**  Let  $\phi = 2x + 3y + 6z - 12$ . The gradient is:

$$\nabla\phi = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}$$

The magnitude is

$$\begin{aligned}\sqrt{2^2 + 3^2 + 6^2} &= \sqrt{49} = 7. \\ \hat{\mathbf{n}} &= \frac{2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}}{7} = \frac{1}{7}(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}})\end{aligned}$$

**Step 3: Choose a Projection Plane** We project the surface  $S$  onto the  $xy$ -plane. Let's call this 2D projected region  $S_1$ . The projection formula for  $dS$  on the  $xy$ -plane is

$$\begin{aligned}dS &= \frac{dx dy}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}|}. \\ \hat{\mathbf{n}} \cdot \hat{\mathbf{k}} &= \frac{1}{7}(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}) \cdot \hat{\mathbf{k}} = \frac{6}{7}\end{aligned}$$

Thus,

$$dS = \frac{dx dy}{6/7} = \frac{7}{6} dx dy.$$

**Step 4: Establish Limits of Integration for  $S_1$**  On the  $xy$ -plane ( $z = 0$ ), the plane equation becomes  $2x + 3y = 12$ . This line bounds the triangular region  $S_1$ . -  $x$  varies from 0 to 6. - For a given  $x$ ,  $y$  varies from 0 to the line

$$y = \frac{12 - 2x}{3}.$$

**\*\*Step 5: Compute**

$$\mathbf{F} \cdot \hat{\mathbf{n}}$$

\*\*

$$\mathbf{F} \cdot \hat{\mathbf{n}} = (6z\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + y\hat{\mathbf{k}}) \cdot \frac{1}{7}(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}) = \frac{1}{7}(12z - 12 + 6y)$$

Because we are integrating with respect to  $x$  and  $y$ , we must eliminate  $z$ . Using the plane equation  $6z = 12 - 2x - 3y$ , we substitute  $12z$ :

$$12z = 24 - 4x - 6y$$

$$\mathbf{F} \cdot \hat{\mathbf{n}} = \frac{1}{7}(24 - 4x - 6y - 12 + 6y) = \frac{1}{7}(12 - 4x)$$

**Step 6: Evaluate the Double Integral** Substitute everything into the integral:

$$\begin{aligned} \iint_{S_1} \left[ \frac{1}{7}(12 - 4x) \right] \left( \frac{7}{6} dx dy \right) &= \int_0^6 \int_0^{\frac{12-2x}{3}} \frac{1}{6}(12 - 4x) dy dx \\ &= \int_0^6 \int_0^{\frac{12-2x}{3}} \left( 2 - \frac{2}{3}x \right) dy dx = \frac{1}{3} \int_0^6 \int_0^{\frac{12-2x}{3}} (6 - 2x) dy dx \end{aligned}$$

Integrate with respect to  $y$  first:

$$\begin{aligned} \frac{1}{3} \int_0^6 (6 - 2x) [y]_0^{\frac{12-2x}{3}} dx &= \frac{1}{3} \int_0^6 (6 - 2x) \left( \frac{12 - 2x}{3} \right) dx \\ &= \frac{1}{9} \int_0^6 4(3 - x)(6 - x) dx = \frac{4}{9} \int_0^6 (18 - 9x + x^2) dx \end{aligned}$$

Integrate with respect to  $x$ :

$$\frac{4}{9} \left[ 18x - \frac{9x^2}{2} + \frac{x^3}{3} \right]_0^6 = \frac{4}{9} [108 - 162 + 72] = \frac{4}{9} [18] = 8$$

Final Answer: 8

## 7.5 Key Takeaways

- Surface integrals require calculating a unit normal vector (

$$\hat{\mathbf{n}} = \nabla\phi/|\nabla\phi|$$

) and computing the dot product

$$\mathbf{F} \cdot \hat{\mathbf{n}}.$$

- To evaluate the integral, you **must project the 3D surface onto a 2D plane**. This transforms the  $dS$  element and defines the integration limits.
- Once projected, ensure the entire integrand (

$$\mathbf{F} \cdot \hat{\mathbf{n}}$$

) is expressed strictly in terms of the two projected variables.

## 7.6 What Comes Next

## 8 Lecture 08: Vector Calculus - Concept of Surface and Volume Integral (Part 2)

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 08 of 14 **Video:** <https://www.youtube.com/watch?v=cHe58sjHmvg>

### 8.1 Prerequisites

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS$$

And solved a problem by projecting the 3D surface onto the  $xy$ -plane. In this lecture, we will demonstrate that the choice of projection plane does not alter the final result, and we will introduce **Spherical Coordinates** to drastically simplify integration over spherical surfaces.

### 8.2 Solved Problems

#### 8.2.1 Problem 1 (Revisited): Projection on the-plane

**Question:** If

$$\mathbf{F} = 6z\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + y\hat{\mathbf{k}},$$

Evaluate

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS,$$

Where  $S$  is the portion of the plane  $2x + 3y + 6z = 12$  in the first octant.

**Solution:**

**Step 1: The Unit Normal Vector** We already calculated  $\hat{\mathbf{n}}$ :

$$\hat{\mathbf{n}} = \frac{1}{7}(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}})$$

**Step 2: Projection on the  $yz$ -plane** When projecting onto the  $yz$ -plane, the area element is:

$$dS = \frac{dy dz}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{i}}|}$$

Compute the dot product:

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{i}} = \frac{2}{7}.$$

So,

$$dS = \frac{dy dz}{2/7} = \frac{7}{2} dy dz.$$

**Step 3: Establish Limits on the  $yz$ -plane** Set  $x = 0$  in the plane equation:  $3y + 6z = 12 \implies y + 2z = 4$ . This line forms a triangle with the  $y$  and  $z$  axes. -  $y$  varies from 0 to 4 -  $z$  varies from 0 to the line

$$z = 2 - \frac{y}{2}$$

**Step 4: Compute the Integrand** We already know

$$\mathbf{F} \cdot \hat{\mathbf{n}} = \frac{1}{7}(12z - 12 + 6y).$$

Because we are integrating with respect to  $y$  and  $z$ , there is no  $x$  variable to eliminate. We can use this directly!

**Step 5: Evaluate the Integral**

$$\iint_S (\mathbf{F} \cdot \hat{\mathbf{n}}) dS = \int_0^4 \int_0^{2-y/2} \frac{1}{7} (12z - 12 + 6y) \left( \frac{7}{2} dz dy \right)$$

Cancel the 7's and divide the terms by 2:

$$= \int_0^4 \int_0^{2-y/2} (6z - 6 + 3y) dz dy$$

Integrating with respect to  $z$  first, then  $y$ , yields 8. The answer is identical to the  $xy$ -plane projection.

### 8.2.2 Problem 2: Surface Integral Over a Sphere

**Question:** Evaluate

$$\iint_S (yz\hat{\mathbf{i}} + zx\hat{\mathbf{j}} + xy\hat{\mathbf{k}}) \cdot d\mathbf{S},$$

Where  $S$  is the surface of the sphere

$$x^2 + y^2 + z^2 = 1$$

in the first octant.

(Note:  $d\mathbf{S}$  in bold implies  $\hat{\mathbf{n}} dS$ )

**Solution:** Let

$$\phi = x^2 + y^2 + z^2 - 1.$$

The gradient is

$$\nabla\phi = 2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} + 2z\hat{\mathbf{k}}.$$

The magnitude is

$$|\nabla\phi| = \sqrt{4(x^2 + y^2 + z^2)} = \sqrt{4(1)} = 2.$$

Thus, the unit normal vector is:

$$\hat{\mathbf{n}} = \frac{2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} + 2z\hat{\mathbf{k}}}{2} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

Compute

$$\mathbf{F} \cdot \hat{\mathbf{n}} :$$

$$(yz\hat{\mathbf{i}} + zx\hat{\mathbf{j}} + xy\hat{\mathbf{k}}) \cdot (x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}) = xyz + xyz + xyz = 3xyz$$

We need to evaluate:

$$I = \iint_S 3xyz dS.$$

We will solve this using two different methods to demonstrate the power of coordinate systems.

**8.2.2.1 Method 1: Cartesian Projection (on-plane)** Project the sphere onto the  $xy$ -plane. The projection is a quarter circle (since we are in the first octant) bounded by

$$x^2 + y^2 = 1.$$

- $x$  varies from 0 to 1.
- $y$  varies from 0 to  $\sqrt{1 - x^2}$ .

The area element is

$$dS = \frac{dx dy}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}|}.$$

Since

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = z$$

, we have

$$dS = \frac{dx dy}{z}.$$

Substitute into the integral:

$$I = \iint_{S_1} 3xyz \left( \frac{dx dy}{z} \right) = 3 \iint_{S_1} xy dx dy$$

Notice how beautifully the  $z$  term cancels out! Let's evaluate:

$$\begin{aligned} I &= 3 \int_0^1 \int_0^{\sqrt{1-x^2}} xy dy dx = 3 \int_0^1 x \left[ \frac{y^2}{2} \right]_0^{\sqrt{1-x^2}} dx \\ &= \frac{3}{2} \int_0^1 x(1-x^2) dx = \frac{3}{2} \int_0^1 (x-x^3) dx \\ &= \frac{3}{2} \left[ \frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = \frac{3}{2} \left( \frac{1}{2} - \frac{1}{4} \right) = \frac{3}{2} \left( \frac{1}{4} \right) = \frac{3}{8} \end{aligned}$$

**8.2.2.2 Method 2: Spherical Polar Coordinates** When integrating over a spherical surface, spherical coordinates simplify the process by removing the need for planar projections entirely. For a unit sphere ( $r = 1$ ): -  $x = \sin \theta \cos \phi$  -  $y = \sin \theta \sin \phi$  -  $z = \cos \theta$  - The area element  $dS = \sin \theta d\theta d\phi$  For the first octant, the angles  $\theta$  and  $\phi$  both range from 0 to  $\pi/2$ . Substitute these into

$$I = \iint_S 3xyz dS :$$

$$I = 3 \int_0^{\pi/2} \int_0^{\pi/2} (\sin \theta \cos \phi)(\sin \theta \sin \phi)(\cos \theta)(\sin \theta d\theta d\phi)$$

Group the terms by variable:

$$I = 3 \int_0^{\pi/2} \int_0^{\pi/2} (\sin^3 \theta \cos \theta)(\sin \phi \cos \phi) d\theta d\phi$$

Because the limits are constants, we can separate the integral into the product of two single-variable integrals:

$$I = 3 \left( \int_0^{\pi/2} \sin^3 \theta \cos \theta d\theta \right) \left( \int_0^{\pi/2} \sin \phi \cos \phi d\phi \right)$$

Using  $u$ -substitution ( $u = \sin \theta, du = \cos \theta d\theta$  and  $v = \sin \phi, dv = \cos \phi d\phi$ ):

$$I = 3 \left[ \frac{\sin^4 \theta}{4} \right]_0^{\pi/2} \left[ \frac{\sin^2 \phi}{2} \right]_0^{\pi/2} = 3 \left( \frac{1}{4} \right) \left( \frac{1}{2} \right) = \frac{3}{8}$$

### 8.3 Key Takeaways

- **Projection invariance:** Projecting a 3D surface onto the  $xy$ ,  $yz$ , or  $zx$  planes will always yield the same final integral value, provided the limits and  $dS$  conversions are done correctly.
- **Spherical coordinates:** When the surface  $S$  is part of a sphere, substituting  $x, y, z$  with spherical coordinates and using

$$dS = r^2 \sin \theta d\theta d\phi$$

eliminates the need to project the surface. The entire integral simply becomes a double integral over the angles  $\theta$  and  $\phi$ .



## 9 Lecture 09: Vector Calculus - Concept of Surface and Volume Integral (Part 3)

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 09 of 14 **Video:** [https://www.youtube.com/watch?v=EWpo\\_cqinOI](https://www.youtube.com/watch?v=EWpo_cqinOI)

### 9.1 Prerequisites

In the preceding lectures, we explored surface integrals for flat planes and spherical surfaces. This lecture builds on those concepts to tackle more complex scenarios: evaluating surface integrals over a **closed multi-face surface** (like a cube) and formally calculating a **Volume Integral** over a 3D solid using triple integration.

### 9.2 Solved Problems

#### 9.2.1 Problem 1: Surface Integral Over a Closed Cube

**Question:** Evaluate

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS,$$

Where

$$\mathbf{F} = 4xz\hat{\mathbf{i}} - y^2\hat{\mathbf{j}} + yz\hat{\mathbf{k}}$$

and  $S$  is the closed surface of the cube bounded by the planes  $x = 0, x = a, y = 0, y = a, z = 0, z = a$ .

**Solution:** The closed surface  $S$  of a cube consists of 6 distinct flat faces. To evaluate the surface integral over  $S$ , we must evaluate the integral over each of the 6 faces separately and sum the results.

Since the faces lie on or parallel to the coordinate planes, we don't need to compute gradients. We can directly identify the **outward unit normal vector**  $\hat{\mathbf{n}}$  by observing which direction points away from the interior of the cube. The total integral is:

$$I = I_1 + I_2 + I_3 + I_4 + I_5 + I_6.$$

**1. Front Face ( $x = a$ ) \* Normal:** Outward normal points in the  $+x$  direction, so

$$\hat{\mathbf{n}} = \hat{\mathbf{i}}.$$

- **Area Element:** Projection is on the  $yz$ -plane, so  $dS = dy dz$ .
- **Integrand:**

$$\mathbf{F} \cdot \hat{\mathbf{n}} = 4xz$$

. Since  $x = a$ ,

$$\mathbf{F} \cdot \hat{\mathbf{n}} = 4az.$$

- **Integral:**

$$I_1 = \int_0^a \int_0^a 4az dy dz = 4a \int_0^a z dz \int_0^a dy = 4a \left( \frac{a^2}{2} \right) (a) = 2a^4$$

**2. Back Face ( $x = 0$ ) \* Normal:** Outward normal points in the  $-x$  direction, so

$$\hat{\mathbf{n}} = -\hat{\mathbf{i}}.$$

- **Area Element:**  $dS = dy dz$ .
- **Integrand:**

$$\mathbf{F} \cdot \hat{\mathbf{n}} = -4xz$$

. Since  $x = 0$ ,

$$\mathbf{F} \cdot \hat{\mathbf{n}} = 0.$$

- **Integral:**

$$I_2 = 0$$

3. **Right Face** ( $y = a$ ) \* **Normal:** Outward normal points in the  $+y$  direction, so

$$\hat{\mathbf{n}} = \hat{\mathbf{j}}.$$

- **Area Element:**  $dS = dx \, dz$ .
- **Integrand:**

$$\mathbf{F} \cdot \hat{\mathbf{n}} = -y^2$$

. Since  $y = a$ ,

$$\mathbf{F} \cdot \hat{\mathbf{n}} = -a^2.$$

- **Integral:**

$$I_3 = \int_0^a \int_0^a -a^2 \, dx \, dz = -a^2(a)(a) = -a^4$$

4. **Left Face** ( $y = 0$ ) \* **Normal:** Outward normal points in the  $-y$  direction, so

$$\hat{\mathbf{n}} = -\hat{\mathbf{j}}.$$

- **Area Element:**  $dS = dx \, dz$ .
- **Integrand:**

$$\mathbf{F} \cdot \hat{\mathbf{n}} = y^2$$

. Since  $y = 0$ ,

$$\mathbf{F} \cdot \hat{\mathbf{n}} = 0.$$

- **Integral:**

$$I_4 = 0$$

5. **Top Face** ( $z = a$ ) \* **Normal:** Outward normal points in the  $+z$  direction, so

$$\hat{\mathbf{n}} = \hat{\mathbf{k}}.$$

- **Area Element:**  $dS = dx \, dy$ .
- **Integrand:**

$$\mathbf{F} \cdot \hat{\mathbf{n}} = yz$$

. Since  $z = a$ ,

$$\mathbf{F} \cdot \hat{\mathbf{n}} = ay.$$

- **Integral:**

$$I_5 = \int_0^a \int_0^a ay \, dx \, dy = a \int_0^a y \, dy \int_0^a dx = a \left( \frac{a^2}{2} \right) (a) = \frac{a^4}{2}$$

6. **Bottom Face** ( $z = 0$ ) \* **Normal:** Outward normal points in the  $-z$  direction, so

$$\hat{\mathbf{n}} = -\hat{\mathbf{k}}.$$

- **Area Element:**  $dS = dx \, dy$ .
- **Integrand:**

$$\mathbf{F} \cdot \hat{\mathbf{n}} = -yz$$

. Since  $z = 0$ ,

$$\mathbf{F} \cdot \hat{\mathbf{n}} = 0.$$

- **Integral:**

$$I_6 = 0$$

**Total Integral:** Summing the results from all 6 faces:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS = 2a^4 + 0 - a^4 + 0 + \frac{a^4}{2} + 0 = a^4 + \frac{a^4}{2} = \frac{3a^4}{2}$$

### 9.2.2 Problem 2: Volume Integral Over a Solid Cylinder

**Question:** Evaluate

$$\iiint_V (\nabla \cdot \mathbf{F}) dV$$

where

$$\mathbf{F} = 4x\hat{\mathbf{i}} - 2y^2\hat{\mathbf{j}} + z^2\hat{\mathbf{k}},$$

And  $V$  is the solid region bounded by the cylinder

$$x^2 + y^2 = 4$$

and the planes  $z = 0$  and  $z = 3$ .

**Solution: Step 1: Compute the Integrand ( $\nabla \cdot \mathbf{F}$ )** Calculate the divergence of the vector field:

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(4x) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2) = 4 - 4y + 2z$$

**Step 2: Determine Limits of Integration** The volume  $V$  is a solid cylinder. The height is bounded by the planes  $z = 0$  and  $z = 3$ . The base of the cylinder on the  $xy$ -plane is the circle

$$x^2 + y^2 = 4$$

(radius  $r = 2$ ). - **Limits for  $z$ :** Direct boundary conditions give  $z = 0$  to  $z = 3$ . - **Limits for  $x$  and  $y$ :** From the base circle

$$x^2 + y^2 = 4$$

,  $x$  varies from  $-2$  to  $2$ . For a given  $x$ ,  $y$  varies from the lower half of the circle to the upper half, so  $y$  goes from  $-\sqrt{4-x^2}$  to  $\sqrt{4-x^2}$ .

**Step 3: Setup and Evaluate the Triple Integral**

$$I = \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^3 (4 - 4y + 2z) dz dy dx$$

**Inner Integration (with respect to  $z$ ):**

$$\int_0^3 (4 - 4y + 2z) dz = [4z - 4yz + z^2]_0^3 = (12 - 12y + 9) - 0 = 21 - 12y = 3(7 - 4y)$$

**Middle Integration (with respect to  $y$ ):** Substitute the result back:

$$\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 3(7 - 4y) dy = 3 [7y - 2y^2]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}$$

Apply the upper and lower limits:

$$= 3 \left[ \left( 7\sqrt{4-x^2} - 2(4-x^2) \right) - \left( 7(-\sqrt{4-x^2}) - 2(4-x^2) \right) \right]$$

The squared terms cancel out:

$$= 3 [14\sqrt{4-x^2}] = 42\sqrt{4-x^2}$$

**Outer Integration (with respect to  $x$ ):** Substitute the result back:

$$I = \int_{-2}^2 42\sqrt{4-x^2} dx$$

Use the standard integral formula:

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left( \frac{x}{a} \right)$$

. Here  $a = 2$ , so

$$a^2 = 4.$$

$$I = 42 \left[ \frac{x}{2} \sqrt{4 - x^2} + 2 \sin^{-1} \left( \frac{x}{2} \right) \right]_{-2}^2$$

When evaluating at  $x = 2$  and  $x = -2$ , the  $\sqrt{4 - x^2}$  term becomes 0.

$$= 42 \left[ (0 + 2 \sin^{-1}(1)) - (0 + 2 \sin^{-1}(-1)) \right]$$

Since

$$\sin^{-1}(1) = \frac{\pi}{2}$$

and

$$\sin^{-1}(-1) = -\frac{\pi}{2} :$$

$$= 42 \left[ 2 \left( \frac{\pi}{2} \right) - 2 \left( -\frac{\pi}{2} \right) \right] = 42[\pi - (-\pi)] = 42(2\pi) = 84\pi$$

Final Answer:  $84\pi$

### 9.3 Key Takeaways

- **Closed Surfaces:** When evaluating a surface integral over a closed shape like a cube, you must evaluate the integral over each individual face and sum the results. The normal vector  $\hat{\mathbf{n}}$  must *always* point outward from the enclosed volume.
- **Volume Integration:** To perform a volume integral over a symmetric solid like a cylinder, start by identifying the constant height bounds for  $z$ . Then, project the solid onto the  $xy$ -plane to determine the limits for  $x$  and  $y$ , which often involves circle boundaries.
- **Sequential Integration:** Always integrate variables with functional limits (like  $y = f(x)$ ) before variables with constant limits.

### 9.4 What Comes Next

With a firm understanding of Line, Surface, and Volume Integrals, we are now fully equipped to tackle the monumental theorems of Vector Calculus: **Gauss' Divergence Theorem**, **Stokes' Theorem**, and **Green's Theorem**.

## 10 Lecture 10: Vector Calculus - Concept of Stokes' Theorem (Part 1)

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 10 of 14 **Video:** <https://www.youtube.com/watch?v=uuWw4hG0s3Q>

### 10.1 Prerequisites

We have extensively studied both Line Integrals

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

And Surface Integrals (

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS$$

). We are now ready to establish a powerful relationship between these two types of integrals using **Stokes' Theorem**.

### 10.2 Stokes' Theorem

#### 10.2.1 Statement

Let  $S$  be an open, two-sided surface bounded by a closed, non-intersecting curve  $C$ . If a vector point function  $\mathbf{F}$  is continuous and differentiable over a region containing  $S$  and  $C$ , then:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$$

Where:  $\oint_C$  denotes the line integral around the closed boundary curve  $C$ .  $\iint_S$  denotes the surface integral over the surface  $S$  bounded by  $C$ .  $\nabla \times \mathbf{F}$  is the curl of the vector field  $\mathbf{F}$ .  $\hat{\mathbf{n}}$  is the outward unit normal vector to the surface  $S$ .

#### 10.2.2 Orientation (The Right-Hand Screw Rule)

The theorem requires a strict convention between the direction you traverse the boundary curve  $C$  and the direction of the outward unit normal  $\hat{\mathbf{n}}$ .

If you traverse the boundary curve  $C$  in the **anticlockwise (positive) direction**, and you curl the fingers of your right hand in that direction, your thumb points in the direction of the outward unit normal  $\hat{\mathbf{n}}$ . For example, if a surface lies flat in the  $xy$ -plane and you traverse its boundary anticlockwise, the outward normal points "upward" along the positive  $z$ -axis (

$$\hat{\mathbf{n}} = \hat{\mathbf{k}}$$

).

#### 10.2.3 Why is this Theorem Useful?

Stokes' Theorem allows you to swap a difficult line integral for an easier surface integral, or vice versa. For example, integrating a vector field along a rectangular boundary requires calculating 4 separate line integrals and summing them up. Using Stokes' theorem, you can simply calculate the curl of the vector field and evaluate a *single* surface integral over the flat rectangular area.

## 10.3 Solved Problems

### 10.3.1 Problem 1: Verifying Stokes' Theorem on a Rectangle

**Question:** Verify Stokes' Theorem for

$$\mathbf{F} = (x^2 + y^2)\hat{\mathbf{i}} - 2xy\hat{\mathbf{j}}$$

taken round the rectangle bounded by  $x = \pm a, y = 0, y = b$ .

**Solution:** To “verify” the theorem, we must independently calculate both the Left-Hand Side (Line Integral) and the Right-Hand Side (Surface Integral) and prove they yield the exact same result.

**10.3.1.1 Part 1: Evaluating the Line Integral (LHS)** Let's define the rectangular boundary  $C$  in the  $xy$ -plane with vertices:  $C(-a, 0)$ ,  $B(a, 0)$ ,  $A(a, b)$ , and  $D(-a, b)$ . We will traverse this boundary in the **anticlockwise direction**. This divides the boundary into 4 line segments: 1.  $C_3$  (**Bottom**): From  $C(-a, 0)$  to  $B(a, 0)$ . Here,  $y = 0 \implies dy = 0$ .  $x$  goes from  $-a$  to  $a$ . 2.  $C_4$  (**Right**): From  $B(a, 0)$  to  $A(a, b)$ . Here,  $x = a \implies dx = 0$ .  $y$  goes from  $0$  to  $b$ . 3.  $C_1$  (**Top**): From  $A(a, b)$  to  $D(-a, b)$ . Here,  $y = b \implies dy = 0$ .  $x$  goes from  $a$  to  $-a$ . 4.  $C_2$  (**Left**): From  $D(-a, b)$  to  $C(-a, 0)$ . Here,  $x = -a \implies dx = 0$ .  $y$  goes from  $b$  to  $0$ . The general line integrand is

$$\mathbf{F} \cdot d\mathbf{r} = (x^2 + y^2)dx - 2xy dy$$

. Let's evaluate this for each segment:

**On  $C_3$  (Bottom):**

$$I_3 = \int_{-a}^a (x^2 + 0) dx = \left[ \frac{x^3}{3} \right]_{-a}^a = \frac{a^3}{3} - \left( -\frac{a^3}{3} \right) = \frac{2a^3}{3}$$

**On  $C_4$  (Right):**

$$I_4 = \int_0^b -2(a)y dy = -2a \left[ \frac{y^2}{2} \right]_0^b = -ab^2$$

**On  $C_1$  (Top):**

$$I_1 = \int_a^{-a} (x^2 + b^2) dx = \left[ \frac{x^3}{3} + b^2x \right]_a^{-a} = \left( -\frac{a^3}{3} - ab^2 \right) - \left( \frac{a^3}{3} + ab^2 \right) = -\frac{2a^3}{3} - 2ab^2$$

**On  $C_2$  (Left):**

$$I_2 = \int_b^0 -2(-a)y dy = 2a \int_b^0 y dy = 2a \left[ \frac{y^2}{2} \right]_b^0 = 2a \left( 0 - \frac{b^2}{2} \right) = -ab^2$$

Summing the four line integrals gives the total LHS:

$$\text{LHS} = I_1 + I_2 + I_3 + I_4 = \left( -\frac{2a^3}{3} - 2ab^2 \right) - ab^2 + \left( \frac{2a^3}{3} \right) - ab^2 = -4ab^2$$

**10.3.1.2 Part 2: Evaluating the Surface Integral (RHS)** We need to compute

$$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$$

over the rectangular area  $S$ .

First, compute the curl  $(\nabla \times \mathbf{F})$ :

$$\nabla \times \mathbf{F} = \det \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + y^2 & -2xy & 0 \end{vmatrix}$$

$$\begin{aligned}
&= \hat{\mathbf{i}}(0 - 0) - \hat{\mathbf{j}}(0 - 0) + \hat{\mathbf{k}} \left( \frac{\partial}{\partial x}(-2xy) - \frac{\partial}{\partial y}(x^2 + y^2) \right) \\
&= \hat{\mathbf{k}}(-2y - 2y) = -4y\hat{\mathbf{k}}
\end{aligned}$$

Because the surface lies flat in the  $xy$ -plane and we traversed the boundary anticlockwise, the outward unit normal points along the positive  $z$ -axis:

$$\hat{\mathbf{n}} = \hat{\mathbf{k}}.$$

$$(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = (-4y\hat{\mathbf{k}}) \cdot \hat{\mathbf{k}} = -4y$$

Now, set up the double integral over the rectangular bounds ( $x$  from  $-a$  to  $a$ , and  $y$  from  $0$  to  $b$ ):

$$\text{RHS} = \int_{-a}^a \int_0^b -4y \, dy \, dx$$

Integrate with respect to  $y$ :

$$\int_0^b -4y \, dy = -4 \left[ \frac{y^2}{2} \right]_0^b = -2b^2$$

Integrate the result with respect to  $x$ :

$$\int_{-a}^a -2b^2 \, dx = -2b^2 [x]_{-a}^a = -2b^2(a - (-a)) = -2b^2(2a) = -4ab^2$$

**10.3.1.3 Conclusion** We have independently evaluated both sides:

$$\text{LHS} = -4ab^2 \quad \text{and} \quad \text{RHS} = -4ab^2$$

Since  $\text{LHS} = \text{RHS}$ , Stokes' Theorem is strictly verified. Notice how much faster the surface integral approach (RHS) was compared to computing four separate line integrals (LHS).

## 10.4 What Comes Next

## 11 Lecture 11: Vector Calculus - Concept of Stokes' Theorem (Part 2)

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 11 of 14 **Video:** <https://www.youtube.com/watch?v=P74t2Ix5NZI>

### 11.1 Prerequisites

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$$

For a rectangular boundary on a 2D plane. In this lecture, we will verify the theorem for a circular boundary and, most importantly, for a curved 3D surface (a hemisphere) to show that the theorem holds regardless of the surface geometry.

### 11.2 Solved Problems

#### 11.2.1 Problem 1: Verifying Stokes' Theorem on a Unit Circle

**Question:** Verify Stokes' theorem for the function

$$\mathbf{F} = (x + 2y)\hat{\mathbf{i}} + (y + 3x)\hat{\mathbf{j}}$$

where  $C$  is the unit circle in the  $xy$ -plane.

**Solution:** A unit circle in the  $xy$ -plane has the equation

$$x^2 + y^2 = 1$$

and lies at  $z = 0$ .

**11.2.1.1 Part 1: Evaluating the Line Integral (LHS)** Since  $z = 0$ ,  $dz = 0$ . The line integral becomes:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C [(x + 2y)dx + (y + 3x)dy]$$

To evaluate this around the circular boundary, we convert to parametric coordinates. Let  $\phi$  be the angle parameter from 0 to  $2\pi$ : \*  $x = \cos \phi \implies dx = -\sin \phi d\phi$  \*  $y = \sin \phi \implies dy = \cos \phi d\phi$

Substitute these into the integral:

$$\begin{aligned} & \int_0^{2\pi} [(\cos \phi + 2 \sin \phi)(-\sin \phi) + (\sin \phi + 3 \cos \phi)(\cos \phi)] d\phi \\ &= \int_0^{2\pi} (-\sin \phi \cos \phi - 2 \sin^2 \phi + \sin \phi \cos \phi + 3 \cos^2 \phi) d\phi \end{aligned}$$

The  $-\sin \phi \cos \phi$  terms cancel:

$$= \int_0^{2\pi} (3 \cos^2 \phi - 2 \sin^2 \phi) d\phi$$

Using the identity

$$\begin{aligned} & \sin^2 \phi = 1 - \cos^2 \phi : \\ &= \int_0^{2\pi} (3 \cos^2 \phi - 2(1 - \cos^2 \phi)) d\phi = \int_0^{2\pi} (5 \cos^2 \phi - 2) d\phi \end{aligned}$$

Convert using the double angle formula

$$\cos^2 \phi = \frac{1 + \cos 2\phi}{2} :$$

$$= \int_0^{2\pi} \left( \frac{5}{2}(1 + \cos 2\phi) - 2 \right) d\phi = \int_0^{2\pi} \left( \frac{1}{2} + \frac{5}{2} \cos 2\phi \right) d\phi$$

Integrate:

$$\left[ \frac{\phi}{2} + \frac{5}{4} \sin 2\phi \right]_0^{2\pi} = \left( \frac{2\pi}{2} + 0 \right) - (0 + 0) = \pi$$

So, LHS =  $\pi$ .

#### 11.2.1.2 Part 2: Evaluating the Surface Integral (RHS) We evaluate

$$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$$

over the flat unit disk bounded by the circle. First, compute the curl:

$$\nabla \times \mathbf{F} = \det \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+2y & y+3x & 0 \end{vmatrix} = \hat{\mathbf{k}} \left( \frac{\partial}{\partial x}(y+3x) - \frac{\partial}{\partial y}(x+2y) \right) = \hat{\mathbf{k}}(3-2) = \hat{\mathbf{k}}$$

Since the surface is flat in the  $xy$ -plane, the outward unit normal is

$$\hat{\mathbf{n}} = \hat{\mathbf{k}}.$$

$$(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 1$$

The integral becomes:

$$\iint_S 1 dS = \text{Area}(S)$$

Since  $S$  is a unit circle ( $r = 1$ ), its area is

$$\pi r^2 = \pi.$$

So, RHS =  $\pi$ . **LHS = RHS**, hence verified.

#### 11.2.2 Problem 2: Verifying Stokes' Theorem on a Hemisphere

**Question:** Verify Stokes' theorem for

$$\mathbf{F} = y\hat{\mathbf{i}} + z\hat{\mathbf{j}} + x\hat{\mathbf{k}},$$

Where  $S$  is the upper hemisphere

$$x^2 + y^2 + z^2 = 1 \text{ for } z \geq 0$$

, and  $C$  is its boundary.

**Solution:** The surface  $S$  is a 3D curved hemisphere dome. The boundary  $C$  is the base ring of this dome lying in the  $xy$ -plane, which is exactly the unit circle

$$x^2 + y^2 = 1$$

at  $z = 0$ .

**11.2.2.1 Part 1: Evaluating the Line Integral (LHS)** Since the boundary  $C$  lies entirely in the  $xy$ -plane,  $z = 0$  and  $dz = 0$ . The integral

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C (y dx + z dy + x dz)$$

simplifies to:

$$\oint_C y dx$$

Using parametric equations for the unit circle  $x = \cos \phi \implies dx = -\sin \phi d\phi$ , and  $y = \sin \phi$ :

$$\begin{aligned} \int_0^{2\pi} \sin \phi (-\sin \phi) d\phi &= - \int_0^{2\pi} \sin^2 \phi d\phi = - \int_0^{2\pi} \frac{1 - \cos 2\phi}{2} d\phi \\ &= -\frac{1}{2} \left[ \phi - \frac{\sin 2\phi}{2} \right]_0^{2\pi} = -\frac{1}{2}(2\pi - 0) = -\pi \end{aligned}$$

So, LHS =  $-\pi$ .

### 11.2.2.2 Part 2: Evaluating the Surface Integral (RHS) We evaluate

$$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$$

over the **curved hemispherical surface**. Compute the curl:

$$\nabla \times \mathbf{F} = \det \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} = \hat{\mathbf{i}}(0 - 1) - \hat{\mathbf{j}}(1 - 0) + \hat{\mathbf{k}}(0 - 1) = -\hat{\mathbf{i}} - \hat{\mathbf{j}} - \hat{\mathbf{k}}$$

Find the unit normal vector  $\hat{\mathbf{n}}$  for the sphere

$$g(x, y, z) = x^2 + y^2 + z^2 - 1 = 0 :$$

$$\nabla g = 2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} + 2z\hat{\mathbf{k}} \implies \hat{\mathbf{n}} = \frac{\nabla g}{|\nabla g|} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

Compute the dot product:

$$(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = (-\hat{\mathbf{i}} - \hat{\mathbf{j}} - \hat{\mathbf{k}}) \cdot (x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}) = -(x + y + z)$$

We convert this to spherical polar coordinates to integrate over the curved surface ( $r = 1$ ): \*  $x = \sin \theta \cos \phi$  \*  $y = \sin \theta \sin \phi$  \*  $z = \cos \theta$  \*  $dS = \sin \theta d\theta d\phi$  \* Limits:  $\phi$  from 0 to  $2\pi$  (full circle),  $\theta$  from 0 to  $\pi/2$  (upper hemisphere only).

Set up the integral:

$$\text{RHS} = \int_0^{2\pi} \int_0^{\pi/2} -(\sin \theta \cos \phi + \sin \theta \sin \phi + \cos \theta) \sin \theta d\theta d\phi$$

Integrate with respect to  $\phi$  first. Over  $[0, 2\pi]$ , the integral of  $\cos \phi$  is 0, and the integral of  $\sin \phi$  is 0. The only term that survives is the  $\cos \theta$  term:

$$\int_0^{2\pi} \cos \theta \sin \theta d\phi = 2\pi \cos \theta \sin \theta$$

Now integrate with respect to  $\theta$ :

$$\begin{aligned} - \int_0^{\pi/2} 2\pi \cos \theta \sin \theta d\theta &= -\pi \int_0^{\pi/2} 2 \sin \theta \cos \theta d\theta = -\pi \int_0^{\pi/2} \sin 2\theta d\theta \\ &= -\pi \left[ -\frac{\cos 2\theta}{2} \right]_0^{\pi/2} = \frac{\pi}{2} (\cos \pi - \cos 0) = \frac{\pi}{2} (-1 - 1) = -\pi \end{aligned}$$

So, RHS =  $-\pi$ . **LHS = RHS**, hence verified.

### 11.2.3 Practice Application

**Question:** Evaluate

$$\oint_C (xy \, dx + xy^2 \, dy)$$

taken round the positively oriented square with vertices  $(1, 0)$ ,  $(0, 1)$ ,  $(-1, 0)$ , and  $(0, -1)$  by using Stokes' theorem.

**Solution Approach:** Instead of evaluating 4 separate line integrals for the square boundary, we use Stokes' Theorem to evaluate the surface integral: 1.

$$\mathbf{F} = xy\hat{\mathbf{i}} + xy^2\hat{\mathbf{j}}.$$

2.

$$\nabla \times \mathbf{F} = (y^2 - x)\hat{\mathbf{k}}.$$

3.

$$\hat{\mathbf{n}} = \hat{\mathbf{k}}$$

(square lies in  $xy$ -plane). 4. Integral:  $\iint_S (y^2 - x) \, dx \, dy$ . By symmetry, the  $\iint_S x \, dx \, dy$  component over the centered square is 0. The remaining integral

$$4 \int_0^1 \int_0^{1-x} y^2 \, dy \, dx$$

evaluates to exactly  $\frac{1}{3}$ .

## 11.3 What Comes Next

We have completed our study of Stokes' Theorem. In the next phase, we will move on to the **Gauss Divergence Theorem**, which relates surface integrals to volume integrals.

## 12 Lecture 12: Vector Calculus - Concept of Gauss Divergence Theorem (Part 1)

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 12 of 14 **Video:** <https://www.youtube.com/watch?v=vZas0kR9cUY>

### 12.1 Prerequisites

We have already seen how **Stokes' Theorem** relates line integration to surface integration. Now, we introduce the **Gauss Divergence Theorem**, which provides a profound connection between **surface integrals** and **volume integrals**. A strong foundation in double and triple integration bounds is essential here.

### 12.2 Gauss Divergence Theorem

#### 12.2.1 Statement

Let  $V$  be a solid volume bounded by a closed surface  $S$ . If  $\mathbf{F}$  is a continuously differentiable vector point function, then the surface integral of  $\mathbf{F}$  over  $S$  is equal to the volume integral of the divergence of  $\mathbf{F}$  over  $V$ :

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS = \iiint_V (\nabla \cdot \mathbf{F}) \, dV$$

Where:  $\ast \iint_S$  denotes the surface integral over the *closed* surface  $S$ .  $\ast \iiint_V$  denotes the triple (volume) integral over the enclosed space  $V$ .  $\ast \hat{\mathbf{n}}$  is the outward-pointing unit normal vector to the surface  $S$ .  $\ast \nabla \cdot \mathbf{F}$  is the divergence (a scalar) of the vector field  $\mathbf{F}$ .

#### 12.2.2 Why is this Theorem Useful?

Evaluating a surface integral over a fully closed shape (like a cube, sphere, or tetrahedron) requires calculating the flux across every single bounding face separately. The Gauss Divergence Theorem allows us to bypass this tedious calculation entirely. Instead of integrating over multiple external surfaces, we compute the divergence of the vector field and evaluate a single triple integral over the solid interior volume.

### 12.3 Solved Problems

#### 12.3.1 Problem 1: Integration over a Tetrahedron

**Question:** Evaluate the surface integral

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS,$$

Where

$$\mathbf{F} = (x + y^2)\hat{\mathbf{i}} - 2x\hat{\mathbf{j}} + 2yz\hat{\mathbf{k}},$$

And  $S$  is the closed surface bounded by the coordinate planes ( $x = 0, y = 0, z = 0$ ) and the plane  $2x + y + 2z = 6$ .

**Solution:** The surface  $S$  is a tetrahedron. Evaluating this directly would require 4 separate double integrals (one for each of the 4 triangular faces). Instead, we will use the Gauss Divergence Theorem to evaluate a single volume integral over the interior space  $V$ .

**Step 1: Compute the Divergence**

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x + y^2) + \frac{\partial}{\partial y}(-2x) + \frac{\partial}{\partial z}(2yz) = 1 + 0 + 2y = 1 + 2y$$

The volume integral is:  $\iiint_V (1 + 2y) \, dV$ .

**Step 2: Determine Limits of Integration** We find the intercepts of the plane  $2x + y + 2z = 6$ : \*  $x$ -intercept (set  $y = z = 0$ ):  $2x = 6 \implies x = 3$  \*  $y$ -intercept (set  $x = z = 0$ ):  $y = 6$  \*  $z$ -intercept (set  $x = y = 0$ ):  $2z = 6 \implies z = 3$

The limits are established by working from the top down to the base projection: 1.  **$z$ -limits:** From the floor ( $z = 0$ ) to the slanted roof (

$$z = \frac{6 - 2x - y}{2}$$

). 2.  **$y$ -limits:** On the floor projection ( $z = 0$ ), the plane becomes the line  $2x + y = 6 \implies y = 6 - 2x$ . So,  $y$  goes from 0 to  $6 - 2x$ . 3.  **$x$ -limits:** From the origin to the  $x$ -intercept ( $x = 3$ ). So,  $x$  goes from 0 to 3.

**Step 3: Evaluate the Triple Integral**

$$I = \int_0^3 \int_0^{6-2x} \int_0^{\frac{6-2x-y}{2}} (1+2y) dz dy dx$$

**Inner Integral (w.r.t  $z$ ):**

$$\int_0^{\frac{6-2x-y}{2}} (1+2y) dz = (1+2y)[z]_0^{\frac{6-2x-y}{2}} = (1+2y) \left( \frac{6-2x-y}{2} \right)$$

Expanding this expression gives:

$$\frac{1}{2}(6-2x-y+12y-4xy-2y^2) = \frac{1}{2}(6-2x+11y-4xy-2y^2)$$

**Middle Integral (w.r.t  $y$ ):**

$$\frac{1}{2} \int_0^{6-2x} (6-2x+11y-4xy-2y^2) dy = \frac{1}{2} \left[ (6-2x)y + \frac{11}{2}y^2 - 2xy^2 - \frac{2}{3}y^3 \right]_0^{6-2x}$$

Substitute the upper limit  $y = 6 - 2x$ :

$$= \frac{1}{2} \left[ (6-2x)^2 + \frac{11}{2}(6-2x)^2 - 2x(6-2x)^2 - \frac{2}{3}(6-2x)^3 \right]$$

Factor out  $(6-2x)^2$ :

$$= \frac{1}{2}(6-2x)^2 \left( 1 + \frac{11}{2} - 2x - \frac{2}{3}(6-2x) \right) = \dots (\text{after algebraic expansion}) = -\frac{4}{3}x^3 + 13x^2 - 42x + 45$$

**Outer Integral (w.r.t  $x$ ):**

$$\begin{aligned} \int_0^3 \left( -\frac{4}{3}x^3 + 13x^2 - 42x + 45 \right) dx &= \left[ -\frac{1}{3}x^4 + \frac{13}{3}x^3 - 21x^2 + 45x \right]_0^3 \\ &= -\frac{1}{3}(81) + \frac{13}{3}(27) - 21(9) + 45(3) = -27 + 117 - 189 + 135 = 36 \end{aligned}$$

[!NOTE] **Correction:** The original lecture contained an algebraic expansion error here. It incorrectly simplified the polynomial to  $-4x^3$  (missing the division by 3) and subsequently forced a flawed arithmetic addition  $(-81 + 117 - 189 + 135)$  to arrive at the desired final answer of 36. The mathematical steps have been corrected above to ensure strict logical integrity.

**Final Answer:** 36

### 12.3.2 Problem 2: Integration over a Sphere

**Question:** Evaluate the surface integral

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS,$$

Where

$$\mathbf{F} = x\hat{\mathbf{i}} - y\hat{\mathbf{j}} + 2z\hat{\mathbf{k}},$$

And  $S$  is the boundary of the sphere

$$x^2 + y^2 + (z - 1)^2 = 1.$$

**Solution:** We use Gauss Divergence Theorem to convert this to a volume integral.

**Step 1: Compute Divergence**

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(-y) + \frac{\partial}{\partial z}(2z) = 1 - 1 + 2 = 2$$

**Step 2: Evaluate the Volume Integral** The integral over the spherical volume  $V$  becomes:

$$\iiint_V (\nabla \cdot \mathbf{F}) \, dV = \iiint_V 2 \, dV = 2 \iiint_V dV$$

Notice that  $\iiint_V dV$  is simply the geometric volume of the solid region  $V$ . The region is a sphere with equation

$$x^2 + y^2 + (z - 1)^2 = 1$$

, which has a radius  $R = 1$  and is centered at  $(0, 0, 1)$ .

The geometric volume of a sphere is

$$V = \frac{4}{3}\pi R^3$$

. For  $R = 1$ :

$$V = \frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi$$

Substitute this back into the expression:

$$2 \iiint_V dV = 2(V) = 2\left(\frac{4}{3}\pi\right) = \frac{8}{3}\pi$$

**Final Answer:**  $\frac{8}{3}\pi$

[!NOTE] This problem demonstrates a massive shortcut: if the divergence  $\nabla \cdot \mathbf{F}$  is a constant scalar, you do not need to set up complex integration bounds. You can simply multiply that constant by the known geometric volume formula of the enclosed solid!

## 12.4 What Comes Next

## 13 Lecture 13: Vector Calculus - Concept of Gauss Divergence Theorem (Part 2)

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 13 of 14 **Video:** <https://www.youtube.com/watch?v=IjWYwfuPBKU>

### 13.1 Prerequisites

We previously used the Gauss Divergence Theorem

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS = \iiint_V (\nabla \cdot \mathbf{F}) dV$$

To evaluate surface integrals by converting them into simpler volume integrals. In this lecture, we will perform a full **verification** of the theorem over a composite surface (a cylinder) and explore a neat trick utilizing the center of gravity to solve cone integrals instantly.

### 13.2 Solved Problems

#### 13.2.1 Problem 1: Verifying Gauss Divergence Theorem on a Cylinder

**Question:** Verify the divergence theorem for

$$\mathbf{F} = 4x\hat{\mathbf{i}} - 2y^2\hat{\mathbf{j}} + z^2\hat{\mathbf{k}}$$

taken over the region bounded by

$$x^2 + y^2 = 4$$

,  $z = 0$ , and  $z = 3$ .

**Solution:** The region is a solid cylinder of radius  $R = 2$  and height  $h = 3$ . To verify the theorem, we must evaluate both the volume integral (RHS) and the surface integral (LHS) and ensure they are equal.

##### 13.2.1.1 Part 1: Evaluating the Volume Integral (RHS)

First, calculate the divergence of  $\mathbf{F}$ :

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(4x) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2) = 4 - 4y + 2z$$

The triple integral is

$$\iiint_V (4 - 4y + 2z) dx dy dz.$$

The bounds for the cylinder are: \*  $z$  from 0 to 3. \*  $y$  from  $-\sqrt{4-x^2}$  to  $\sqrt{4-x^2}$  (from the circular base). \*  $x$  from  $-2$  to 2.

Integrate with respect to  $z$ :

$$\int_0^3 (4 - 4y + 2z) dz = [4z - 4yz + z^2]_0^3 = 12 - 12y + 9 = 21 - 12y = 3(7 - 4y)$$

Integrate with respect to  $y$ :

$$3 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (7 - 4y) dy = 3 [7y - 2y^2]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}$$

Because  $y^2$  is an even function, the subtraction of limits causes the  $y^2$  terms to cancel out entirely.

$$= 3 \left( 14\sqrt{4-x^2} \right) = 42\sqrt{4-x^2}$$

Integrate with respect to  $x$ :

$$I_V = \int_{-2}^2 42\sqrt{4-x^2} dx$$

Using the standard integration formula

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left( \frac{x}{a} \right) :$$

$$I_V = 42 \left[ \frac{x}{2} \sqrt{4-x^2} + \frac{4}{2} \sin^{-1} \left( \frac{x}{2} \right) \right]_{-2}^2$$

When evaluated at  $x = \pm 2$ , the square root term becomes 0.

$$= 42 [2 \sin^{-1}(1) - 2 \sin^{-1}(-1)] = 42 \left( 2 \left( \frac{\pi}{2} \right) - 2 \left( -\frac{\pi}{2} \right) \right) = 42(\pi + \pi) = 84\pi$$

So,  $\text{RHS} = 84\pi$ .

**13.2.1.2 Part 2: Evaluating the Surface Integral (LHS)** The closed surface  $S$  consists of 3 parts: the bottom base ( $S_1$ ), the top base ( $S_2$ ), and the curved lateral surface ( $S_3$ ).

**1. Bottom Base ( $S_1$  at  $z = 0$ ):** \* Outward normal:

$$\hat{\mathbf{n}}_1 = -\hat{\mathbf{k}}$$

•

$$\mathbf{F} \cdot \hat{\mathbf{n}}_1 = -z^2.$$

- Since  $z = 0$  on this surface, the integral is exactly 0.

**2. Top Base ( $S_2$  at  $z = 3$ ):** \* Outward normal:

$$\hat{\mathbf{n}}_2 = \hat{\mathbf{k}}$$

•

$$\mathbf{F} \cdot \hat{\mathbf{n}}_2 = z^2.$$

- Since  $z = 3$ , the integrand is

$$3^2 = 9.$$

- The integral is

$$9 \times \text{Area}(S_2) = 9 \times (\pi \cdot 2^2) = 36\pi.$$

**\*\*3. Curved Lateral Surface ( $S_3$  where**

$$x^2 + y^2 = 4$$

**):\*\* \*** Gradient of the surface

$$g = x^2 + y^2 - 4 = 0$$

is

$$\nabla g = 2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}}.$$

- Outward normal

$$\hat{\mathbf{n}}_3 = \frac{2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}}}{\sqrt{4x^2 + 4y^2}} = \frac{x\hat{\mathbf{i}} + y\hat{\mathbf{j}}}{2}.$$

•

$$\mathbf{F} \cdot \hat{\mathbf{n}}_3 = (4x\hat{\mathbf{i}} - 2y^2\hat{\mathbf{j}} + z^2\hat{\mathbf{k}}) \cdot \frac{x\hat{\mathbf{i}} + y\hat{\mathbf{j}}}{2} = 2x^2 - y^3.$$

- Convert to cylindrical coordinates:

$$x = 2 \cos \theta, y = 2 \sin \theta, dS_3 = R d\theta dz = 2 d\theta dz.$$

- Limits:  $\theta \in [0, 2\pi]$  and  $z \in [0, 3]$ .

$$\begin{aligned} \int_0^{2\pi} \int_0^3 (2(2 \cos \theta)^2 - (2 \sin \theta)^3) 2 dz d\theta &= \int_0^{2\pi} \int_0^3 (8 \cos^2 \theta - 8 \sin^3 \theta) 2 dz d\theta \\ &= 48 \int_0^{2\pi} (\cos^2 \theta - \sin^3 \theta) d\theta \end{aligned}$$

Since

$$\int_0^{2\pi} \sin^3 \theta d\theta = 0$$

over a full period, and

$$\begin{aligned} \int_0^{2\pi} \cos^2 \theta d\theta &= \pi :: \\ &= 48(\pi - 0) = 48\pi \end{aligned}$$

**Total Surface Integral:**

$$I_S = I_{S1} + I_{S2} + I_{S3} = 0 + 36\pi + 48\pi = 84\pi$$

**So, LHS =  $84\pi$ .**

Since LHS = RHS =  $84\pi$ , the Gauss Divergence Theorem is fully verified!

### 13.2.2 Problem 2: Center of Gravity Trick for Cones

**Question:** Evaluate

$$\iint_S (x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z^2\hat{\mathbf{k}}) \cdot \hat{\mathbf{n}} dS,$$

Where  $S$  is the closed surface bounded by the cone

$$x^2 + y^2 = z^2$$

and the plane  $z = 1$ .

**Solution:** Using the Divergence Theorem, we convert this to a volume integral.

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z^2) = 1 + 1 + 2z = 2(1 + z)$$

The volume integral is:

$$\iiint_V 2(1 + z) dV = 2 \iiint_V dV + 2 \iiint_V z dV$$

We recognize two geometric properties: 1.  $\iiint_V dV$  is the geometric volume  $V$  of the cone. 2. By definition, the  $z$ -coordinate of the centroid (center of gravity) is

$$\bar{z} = \frac{\iiint_V z dV}{V}.$$

Therefore,

$$\iiint_V z dV = V\bar{z}.$$

Substitute these in:

$$2V + 2V\bar{z} = 2V(1 + \bar{z})$$

Now, we simply use the known geometric formulas for a solid cone: \* The cone has a height  $h = 1$  and a base radius  $R = 1$  (since

$$x^2 + y^2 = z^2$$

at  $z = 1$ ). \* The volume of a cone is

$$V = \frac{1}{3}\pi R^2 h = \frac{1}{3}\pi(1)^2(1) = \frac{1}{3}\pi.$$

- The center of gravity of a solid right circular cone lies on its axis at

$$\bar{z} = \frac{3}{4}h$$

. Since  $h = 1$ ,

$$\bar{z} = \frac{3}{4}.$$

Substitute these known values:

$$\text{Result} = 2 \left( \frac{1}{3}\pi \right) \left( 1 + \frac{3}{4} \right) = \frac{2}{3}\pi \left( \frac{7}{4} \right) = \frac{7}{6}\pi$$

[!TIP] If a divergence evaluates to a linear coordinate (like  $z$ ), you can often solve the integral instantly by utilizing the formula  $V\bar{z}$ , bypassing complex integration setups completely!

### 13.3 What Comes Next

We have completed our study of the Gauss Divergence Theorem. In the final lecture, we will cover **Green's Theorem**, which relates line integrals to double area integrals in the 2D plane.

## 14 Lecture 14: Vector Calculus - Concept of Green's Theorem

**Series:** Vector Calculus by Bhagwan Singh Vishwakarma **Lecture:** 14 of 14 **Video:** <https://www.youtube.com/watch?v=EiWb17ofMnA>

### 14.1 Prerequisites

We have completed our study of Stokes' Theorem (relating 3D line integrals to 3D surface integrals) and Gauss Divergence Theorem (relating 3D surface integrals to 3D volume integrals). We conclude our vector calculus series with **Green's Theorem**, which is essentially a specialized 2D case of Stokes' Theorem. It relates a line integral around a simple closed curve in a 2D plane to a double integral over the flat area enclosed by that curve.

### 14.2 Green's Theorem in a Plane

#### 14.2.1 Statement

If  $P(x, y)$  and  $Q(x, y)$  are continuous functions with continuous partial derivatives in a region  $S$  enclosed by a closed curve  $C$  in the  $xy$ -plane, then:

$$\oint_C (P dx + Q dy) = \iint_S \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

Where: \*  $C$  is traversed in the positive (anticlockwise) direction. \*  $P$  is the expression multiplied by  $dx$ . \*  $Q$  is the expression multiplied by  $dy$ .

#### 14.2.2 Why is this Theorem Useful?

Just like the other theorems, Green's Theorem allows us to bypass tedious boundary integrations. If a boundary  $C$  consists of multiple line segments (like a triangle or a rectangle), evaluating  $\oint_C$  directly requires calculating multiple separate integrals. By applying Green's Theorem, you can simply evaluate a single double integral over the bounded flat region  $S$ .

### 14.3 Solved Problems

#### 14.3.1 Problem 1: Evaluating a Line Integral via Double Integral

**Question:** Evaluate by Green's Theorem the line integral

$$\oint_C (e^{-x} \sin y dx + e^{-x} \cos y dy),$$

Where  $C$  is the rectangle with vertices

$$(0, 0), (\pi, 0), (\pi, \frac{\pi}{2}), (0, \frac{\pi}{2}).$$

**Solution:** We identify the components  $P$  and  $Q$  from the integral form  $\oint_C (P dx + Q dy)$ : \*

$$P = e^{-x} \sin y \implies \frac{\partial P}{\partial y} = e^{-x} \cos y$$

•

$$Q = e^{-x} \cos y \implies \frac{\partial Q}{\partial x} = -e^{-x} \cos y$$

Compute the integrand for the double integral:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -e^{-x} \cos y - e^{-x} \cos y = -2e^{-x} \cos y$$

Set up the double integral over the rectangular region  $S$ . The limits are clearly defined by the vertices:  $x$  varies from 0 to  $\pi$ , and  $y$  varies from 0 to  $\frac{\pi}{2}$ .

$$\iint_S \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \int_0^\pi \int_0^{\frac{\pi}{2}} -2e^{-x} \cos y dy dx$$

Since the limits are constants, we can separate the variables:

$$= -2 \left( \int_0^\pi e^{-x} dx \right) \left( \int_0^{\frac{\pi}{2}} \cos y dy \right)$$

Evaluate the  $y$ -integral:

$$\int_0^{\frac{\pi}{2}} \cos y dy = [\sin y]_0^{\frac{\pi}{2}} = 1 - 0 = 1$$

Evaluate the  $x$ -integral:

$$\int_0^\pi e^{-x} dx = [-e^{-x}]_0^\pi = -e^{-\pi} - (-1) = 1 - e^{-\pi}$$

Multiply them together with the constant  $-2$ :

$$= -2(1 - e^{-\pi})(1) = 2(e^{-\pi} - 1)$$

**Final Answer:**  $2(e^{-\pi} - 1)$

### 14.3.2 Problem 2: Verifying Green's Theorem on a Circle

**Question:** Verify Green's Theorem for the line integral

$$\oint_C (x + 2y) dx + (y + 3x) dy$$

where  $C$  is the unit circle

$$x^2 + y^2 = 1.$$

**Solution:** To verify, we must independently compute both the LHS (Line Integral) and the RHS (Double Integral).

**Part 1: Line Integral (LHS)** Convert the circle

$$x^2 + y^2 = 1$$

to parametric coordinates:  $x = \cos \theta$ ,  $dx = -\sin \theta d\theta$  and  $y = \sin \theta$ ,  $dy = \cos \theta d\theta$ , for  $\theta \in [0, 2\pi]$ .

$$\begin{aligned} \oint_C &= \int_0^{2\pi} [(\cos \theta + 2 \sin \theta)(-\sin \theta) + (\sin \theta + 3 \cos \theta)(\cos \theta)] d\theta \\ &= \int_0^{2\pi} (-\sin \theta \cos \theta - 2 \sin^2 \theta + \sin \theta \cos \theta + 3 \cos^2 \theta) d\theta \end{aligned}$$

The  $\sin \theta \cos \theta$  terms cancel:

$$= \int_0^{2\pi} (3 \cos^2 \theta - 2 \sin^2 \theta) d\theta$$

Using double-angle formulas (

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

and

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

);

$$\begin{aligned} &= \int_0^{2\pi} \left[ 3 \left( \frac{1 + \cos 2\theta}{2} \right) - 2 \left( \frac{1 - \cos 2\theta}{2} \right) \right] d\theta = \int_0^{2\pi} \left( \frac{1}{2} + \frac{5}{2} \cos 2\theta \right) d\theta \\ &= \left[ \frac{\theta}{2} + \frac{5}{4} \sin 2\theta \right]_0^{2\pi} = \frac{2\pi}{2} + 0 = \pi \end{aligned}$$

**LHS** =  $\pi$

**Part 2: Double Integral (RHS)** Identify  $P = x + 2y$  and  $Q = y + 3x$ .

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 3 - 2 = 1$$

The integral becomes:

$$\iint_S 1 \, dx \, dy = \text{Area of region } S$$

Since  $S$  is a circle of radius  $R = 1$ , its area is

$$\pi R^2 = \pi.$$

**RHS** =  $\pi$  Since **LHS** = **RHS** =  $\pi$ , Green's Theorem is verified.

### 14.3.3 Problem 3: Verifying Green's Theorem between Curves

**Question:** Verify Green's Theorem for

$$\oint_C (xy + y^2) \, dx + x^2 \, dy$$

where  $C$  is the closed curve bounded by

$$y = x^2$$

and  $y = x$ .

**Solution:** The intersection points of

$$y = x^2$$

and  $y = x$  are  $(0, 0)$  and  $(1, 1)$ .

**Part 1: Line Integral (LHS)** The boundary  $C$  has two paths traversed in an anticlockwise loop: 1.  $C_1$  (**Lower Curve**): Along

$$y = x^2$$

from  $x = 0$  to  $x = 1$ . ( $dy = 2x \, dx$ )

$$I_1 = \int_0^1 [(x(x^2) + (x^2)^2) + x^2(2x)] \, dx = \int_0^1 (3x^3 + x^4) \, dx = \left[ \frac{3}{4}x^4 + \frac{1}{5}x^5 \right]_0^1 = \frac{19}{20}$$

2.  $C_2$  (**Upper Curve**): Along  $y = x$  from  $x = 1$  to  $x = 0$ . ( $dy = dx$ )

$$I_2 = \int_1^0 [(x^2 + x^2) + x^2] \, dx = \int_1^0 3x^2 \, dx = [x^3]_1^0 = -1$$

Summing the paths:

$$\text{LHS} = \frac{19}{20} - 1 = -\frac{1}{20}.$$

**Part 2: Double Integral (RHS)**

$$P = xy + y^2 \implies \frac{\partial P}{\partial y} = x + 2y.$$

$$Q = x^2 \implies \frac{\partial Q}{\partial x} = 2x.$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - (x + 2y) = x - 2y$$

Set up the bounds: For a given  $x \in [0, 1]$ ,  $y$  ranges from the lower curve (

$$y = x^2$$

) to the upper curve ( $y = x$ ).

$$\begin{aligned} \text{RHS} &= \int_0^1 \int_{x^2}^x (x - 2y) dy dx = \int_0^1 [xy - y^2]_{x^2}^x dx \\ &= \int_0^1 [(x^2 - x^2) - (x^3 - x^4)] dx = \int_0^1 (x^4 - x^3) dx \\ &= \left[ \frac{x^5}{5} - \frac{x^4}{4} \right]_0^1 = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20} \end{aligned}$$

Since **LHS** = **RHS** =  $-\frac{1}{20}$ , Green's Theorem is verified.

## 14.4 Conclusion

This marks the conclusion of our Vector Calculus series! From basic scalar and vector point functions, to the differential operators Gradient, Curl, and Divergence, and finally the monumental theorems of Stokes, Gauss, and Green that unify vector integration. We are now fully equipped to transition into Complex Analysis.